PHYSICS GRADE 10: MECHANICAL PROPERTIES OF MATERIALS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.2 (6 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Physics
Strand	Strand 1.0: Mechanics and Thermal Physics
Sub-Strand	Sub-Strand 1.3: Mechanical Properties of Materials
Total Duration	6 lessons × 40 minutes = 240 minutes total
Sub-Strand Content	– Elasticity: elastic and plastic deformation of materials – Hooke's Law: relationship between force and extension in a spring – Spring constant (k) and its determination from force-extension graphs – Elastic potential energy stored in a stretched spring – Tensile stress, tensile strain, and Young's Modulus – Ductility, brittleness, and malleability as mechanical properties – Practical applications of mechanical properties in everyday materials and engineering
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) describe elasticity and plasticity as mechanical properties of materials, b) state Hooke's Law and apply it to determine the relationship between force and extension, c) determine the spring constant from a force-extension graph, d) calculate elastic potential energy stored in a stretched or compressed spring, e) define tensile stress, tensile strain, and Young's Modulus and solve related problems, f) distinguish between ductile, brittle, and malleable materials with relevant examples, g) appreciate the importance of mechanical properties of materials in engineering, construction, and daily life in Kenya.
Core Competencies	– Communication and Collaboration: the learner works in groups during spring-stretching experiments to share observations, record data on force-extension tables, and present findings to the class, embracing teamwork and joint decision-making. – Critical Thinking and Problem Solving: the learner analyses force-extension graphs, identifies the elastic limit, and applies Hooke's Law to solve quantitative problems involving Kenyan engineering contexts such as suspension bridges and vehicle springs. – Learning to Learn: the learner reflects on and self-corrects experimental technique (e.g., reading a metre rule accurately, avoiding parallax errors) and connects new concepts of stress and strain to prior knowledge of forces from Sub-Strand 1.2. – Digital Literacy: the learner uses digital devices or simulations (e.g., PhET Springs and Masses) to model spring behaviour where physical apparatus is unavailable, effectively accomplishing data collection and graph plotting tasks.
Core Values	– Responsibility: the learner handles laboratory equipment (springs, slotted masses, retort stands) with care, ensures accurate data recording, and takes ownership of their group's experimental results.

	– Integrity: the learner records experimental data honestly, including anomalous results, rather than adjusting figures to fit expectations, thereby upholding scientific honesty. – Respect: the learner appreciates diverse approaches and conclusions presented by different groups during class discussions on material properties and acknowledges the contributions of all group members.
Science & Engineering Practices	– Planning and Carrying Out Investigations (SEP 3): the learner plans and conducts a controlled spring-stretching experiment — selecting variables (force, extension), controlling constants (same spring, same ruler position), and collecting repeated measurements to determine the spring constant. – Analysing and Interpreting Data (SEP 4): the learner plots force-extension graphs, draws lines of best fit, identifies the elastic limit, calculates the gradient (spring constant k), and uses the graph to determine elastic potential energy. – Constructing Explanations (SEP 6): the learner uses Hooke's Law and the concept of Young's Modulus to explain why certain Kenyan construction materials (steel rods in Nairobi high-rises, sisal rope in Rift Valley farms) behave differently under load. – Obtaining, Evaluating, and Communicating Information (SEP 8): the learner evaluates information from data tables and graphs to communicate conclusions about material behaviour, presenting findings clearly with appropriate scientific vocabulary.
Pertinent & Contemporary Issues (PCIs)	– Safety and Security: the learner observes laboratory safety rules during spring-stretching experiments — securing retort stands, wearing safety goggles, and ensuring slotted masses are handled carefully to prevent injury, reinforcing a culture of personal and peer safety. – Life Skills — Critical Thinking: the learner evaluates the suitability of different materials for real-world Kenyan applications (e.g., choosing steel vs. wood for a footbridge over a Rift Valley river) by analysing their mechanical properties, building decision-making skills. – Environmental Education: the learner considers how selecting materials with appropriate mechanical properties reduces material waste and structural failure, linking responsible material choice to environmental sustainability in Kenya's growing construction sector.
Career Connections	– Civil and Structural Engineering: understanding Young's Modulus and tensile strength is fundamental for engineers designing bridges, buildings, and roads across Kenya, including projects such as the Nairobi Expressway. – Materials Science and Metallurgy: knowledge of ductility, brittleness, and elasticity enables materials scientists working with Kenya's manufacturing industries to select and develop appropriate metals and composites. – Orthopaedics and Biomedical Engineering: mechanical properties of materials underpin the design of surgical implants, prosthetic limbs, and orthopaedic supports used in Kenyan hospitals such as Kenyatta National Hospital. – Textile and Agro-processing Industry: understanding elasticity and malleability supports innovation in processing sisal, cotton, and other Kenyan agricultural fibres into ropes, fabrics, and packaging materials. – Automotive Technology: technicians and engineers servicing vehicles in Kenya apply knowledge of spring constants and material fatigue when working on suspension systems and chassis components.
Focus for Lessons	This unit develops learners' understanding of how solid materials respond to applied forces, anchored in the everyday Kenyan experience of elastic and permanent deformation. Learners progress from qualitative observations of stretching rubber bands and springs to quantitative application of Hooke's Law, Young's Modulus, and elastic potential energy, building strong links between experimental evidence and mathematical models. The unit uses a phenomenon-driven, investigation-based approach in which all concepts are developed in the context of explaining a real, puzzling observation about material behaviour.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why do some materials stretch and snap back while others permanently deform or break — and how can we use this knowledge to choose the right material for a safe structure in Kenya? KICD KEY INQUIRY QUESTIONS:

	1. What determines how a material responds when a force is applied to it? 2. How are the mechanical properties of materials applied in engineering and everyday life?
Anchoring Phenomenon	ANCHORING PHENOMENON — The Broken Suspension Bridge Rope at a Rift Valley Farm: A farmer in Kericho uses a hand-made suspension footbridge to cross a small river. The bridge is supported by two thick sisal ropes. One day, after loading a heavy sack of tea leaves (approximately 80 kg) onto the bridge, learners observe a short video clip (or teacher-narrated story with photographs) showing that one rope stretches noticeably and bounces back when the load is removed, while the second — older — rope stretches and then snaps without returning to its original length. What is puzzling: Both ropes look similar and are made of natural fibre, yet they behave completely differently under the same load. Why did one rope return to its original shape while the other did not? Why did the second rope snap at all — what was happening inside the rope material? Can we predict beforehand which rope will snap and which will survive? This puzzle drives the entire unit: learners are compelled to investigate what properties of materials determine how they respond to forces, and how we can measure and predict those properties scientifically.
Storyline Thread	Lesson 1 — Predict & Observe: Elastic vs. Plastic Behaviour Learners are introduced to the anchoring phenomenon (the Kericho bridge ropes). They first predict — individually and in groups — why one rope survived and one snapped, recording ideas on sticky notes for the Driving Question Board (DQB). They then carry out a hands-on exploration: stretching rubber bands, sisal twine, and copper wire with hanging masses, classifying materials as elastic or plastic. Learners record observations and begin building an initial model of what "elasticity" means at the particle level. Lesson 2 — Investigate: Hooke's Law Experiment Learners conduct a structured investigation using a steel spring, slotted masses (0–5 N), a retort stand, and a metre rule. They measure extensions for increasing loads, plot force-extension graphs, identify the elastic limit, and determine the spring constant ($k = F/x$) from the graph gradient. Groups compare k values for different springs and connect findings back to the DQB: "a stiffer spring has a larger k — could this explain why one rope survived?" Learners update their DQB with new questions about quantitative limits. Lesson 3 — Explain: Elastic Potential Energy and the Elastic Limit Learners use their force-extension graphs to calculate elastic potential energy ($E_p = \frac{1}{2}kx^2$) stored in stretched springs, connecting energy storage to the idea of a "safe working load." The teacher introduces the elastic limit formally, and learners explain — using evidence from Lesson 2 — why loading beyond this limit caused the older Kericho rope to fail permanently. Learners apply Hooke's Law quantitatively in worked examples set in Kenyan contexts (vehicle suspension springs, bungee cords at a Nairobi adventure park). Lesson 4 — Extend: Tensile Stress, Strain, and Young's Modulus Learners are introduced to tensile stress ($\sigma = F/A$) and tensile strain ($\varepsilon = \Delta L/L_0$) through guided note-taking and worked examples. They calculate Young's Modulus ($E = \sigma/\varepsilon$) for steel and sisal using provided data sets from a simulated Kenyan materials-testing laboratory. Learners compare E values for different materials (steel, rubber, wood, bone) and revise their particle-level models to explain why a high E value means a material is stiff but not necessarily strong. DQB is updated: "What is the difference between a stiff material and a strong material?" Lesson 5 — Differentiate: Ductility, Brittleness, and Malleability

	Learners examine and classify material samples (copper wire, chalk, plasticine, glass rod, steel nail) by bending, stretching, and compressing them under teacher supervision. They map observations to the formal definitions of ductile, brittle, and malleable, and sketch stress-strain curves for each type. Real Kenyan engineering contexts are analysed: why steel rebar (ductile) is preferred over concrete (brittle) for earthquake-resistant buildings in Nairobi; why copper (malleable) is used for electrical wiring across Kenya. Learners connect each property back to the anchoring phenomenon. Lesson 6 — Model Building & Synthesis: Choosing the Right Material Learners return to the anchoring phenomenon and now construct a full explanatory model: why the new sisal rope behaved elastically (below its elastic limit, high enough Young's Modulus, sufficient cross-sectional area) while the old rope — weakened by use, smaller effective cross-section, reduced elastic limit — exceeded its tensile stress limit and snapped. Groups present their revised models, respond to peer questions, and complete a summative design challenge: recommend a suitable material (with justification using k, E, ductility) for a specific Kenyan infrastructure scenario. DQB is reviewed and all learner questions are addressed.
Supporting Phenomena	– A rubber band used to bundle sukuma wiki at Wakulima Market in Nairobi stretches when pulled and snaps back — but if stretched too far, it remains deformed or breaks. Why does it have a limit? – A steel reinforcement rod (rebar) used in construction of a Nairobi apartment block can be bent into a curve and stays bent permanently, whereas a thin steel spring used in a ballpoint pen returns to its original shape after compression. Why do two steel objects behave so differently? – A glass bottle dropped on a concrete floor in Mombasa shatters into pieces rather than bending or bouncing back. Why do some materials break instantly without any visible stretching beforehand? – The spring inside a matatu passenger seat in Kisumu compresses under the weight of a passenger and returns to its original position when the passenger stands up, lesson after lesson, year after year — yet the spring eventually "wears out" and no longer bounces back. What causes this gradual change?

LESSON 1 (40 minutes): Predict and Observe: Elastic vs Plastic Behaviour — The Kericho Bridge Rope Mystery	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the unit by introducing the anchoring phenomenon and engaging learners in predicting and observing how different materials respond to applied forces, establishing the need for scientific investigation of mechanical properties.
Knowledge	Learners will distinguish between elastic behaviour (material returns to original shape) and plastic behaviour (material does not return to original shape) and identify these behaviours in everyday Kenyan materials including sisal rope, rubber, and copper wire.
Skills	Learners will practise making and recording predictions, carry out a hands-on exploration of material behaviour under load, classify observations using evidence, and collaboratively construct initial explanatory ideas about elasticity at the particle level.
Attitudes	Learners will demonstrate curiosity about puzzling real-world phenomena, show willingness to revise initial ideas based on evidence, and appreciate the relevance of physics to safety and engineering decisions in Kenya.
Key Inquiry Question	What determines how a material responds when a force is applied to it?
Purpose in Storyline	This is the opening lesson that launches the entire unit. It introduces the anchoring phenomenon — the two sisal ropes on the Kericho suspension footbridge — creates genuine puzzlement, activates prior knowledge, and opens the Driving Question Board which will be revisited and updated in every subsequent lesson.
Safety Notes	Learners must not overstretch copper wire near faces — safety goggles should be worn during stretching activities. Masses should be added gently and never exceed 500 g in this exploration. Copper wire ends can be sharp; handle with care. If a rubber band snaps, direct it away from eyes. Teacher must demonstrate safe procedure before learners begin.

B. LESSON OVERVIEW
Learners encounter a short teacher-narrated story with photographs showing two sisal ropes on a Kericho farm suspension footbridge behaving differently under the same 80 kg load — one bounces back, one snaps permanently. This puzzling observation launches individual and group predictions recorded on sticky notes. Learners then carry out a structured hands-on exploration stretching rubber bands, sisal twine, and copper wire with hanging masses, classifying each as elastic or plastic. The lesson closes with the opening of the Driving Question Board and learners sketching an initial particle-level model of what happens inside a material when it is stretched.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Why S-waves only travel in solids] Source: Khan Academy (English	Learners listen to the teacher-narrated Kericho bridge story accompanied by projected photographs: a farmer loading a 80 kg sack of tea leaves onto a sisal rope suspension footbridge over a small river in Kericho	Teacher narrates slowly and expressively: 'Picture yourself on a tea farm in Kericho. You have just harvested 80 kilograms of tea leaves — a full sack. The only way across the river is a small footbridge held up by two thick	Think-Pair-Share anchored to a local Kenyan phenomenon. Sticky note clustering makes prior knowledge visible and creates cognitive dissonance when learners notice that two seemingly identical ropes behave differently,	Teacher reads sticky notes during the pair-share phase to gauge baseline understanding. Look for: Do learners connect material behaviour to physical properties (thickness, age, condition)? Are any learners already using words

- US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Elastic Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	County. Learners see (or hear described) that rope A stretches and bounces back, while rope B stretches permanently and then snaps. Each learner silently writes one prediction on a sticky note answering: 'Why did one rope survive and the other snap?' Learners then share predictions in pairs and agree on a joint statement. Pairs share with the class; teacher records all ideas on the board without evaluating them.	sisal ropes. You load the sack. Rope A stretches — then springs back. Rope B stretches — then snaps. Both ropes look the same. Both are made from natural sisal fibre. Why did they behave so differently?' Teacher then says: 'On your sticky note, write your best prediction. Do not worry about being right — we are scientists making our first guess.' WAIT TIME: Allow 90 seconds of silent individual writing. After pairs share, teacher collects all sticky notes and organises them on the board into rough clusters (e.g. age of rope, thickness, quality of fibre) saying: 'I notice some of you are thinking about the condition of the rope, and others about its size. Let us hold all these ideas — we will test them as the unit goes on.'	driving the need for investigation.	like 'elastic' or 'strength'? This informs pacing and which ideas to highlight when opening the DQB.
Observe Phase 📹 **VIDEO:** Why S-waves only travel in solids Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Elastic Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	In groups of four, learners receive a small kit containing: one rubber band (medium thickness), one 15 cm piece of sisal twine, one 15 cm piece of copper wire, a small plastic cup, and a set of four 50 g masses. Each group follows a structured observation protocol printed on their activity card: (1) Measure and record the resting length of each material. (2) Hook the material between two pencils held horizontally, suspend the plastic cup below, and add masses one at a time up to 200 g. (3) After each addition, observe and record whether the material has changed shape. (4) Remove all masses and record whether the material returns to its original length. Learners use a simple two-column table: 'What happened while loaded' and 'What happened when unloaded'. They classify each material as elastic (returns to	Before learners begin, teacher demonstrates the procedure using a spare rubber band, narrating each step: 'I am adding 50 grams — I observe the band stretches. I remove the mass — it returns. I record: elastic.' Teacher then asks: 'What do you think will happen with the copper wire? Thumbs up for elastic, thumbs sideways for plastic, thumbs down for fracture.' WAIT TIME: 10 seconds for learners to commit to a gesture before revealing results through the activity. During the activity, teacher circulates and uses probing questions: 'What do you notice about the shape of the sisal twine after you removed the masses? What does that tell you?' and 'Where in the Kericho story did you see this same behaviour?' Teacher avoids confirming or denying — redirects back to evidence: 'What does your data	Hands-on direct observation with structured data recording. Connecting each observation back to the Kericho phenomenon builds the evidential base for sensemaking. The two-column table scaffolds comparison thinking and prepares learners for the formal elastic-plastic distinction.	Circulate and check tables. Key diagnostic: Can learners accurately distinguish between what happened under load and what happened after unloading? Common misconception to watch for: learners saying copper wire is elastic because it bent — teacher should prompt: 'Did it return to its original shape when you removed the load?' Groups that finish early are prompted: 'Can you rank the three materials from most elastic to least elastic? How did you decide?'

	original length) or plastic (does not return) or fractured.	say?'		
Explain Phase 📹 **VIDEO:** [Why S-waves only travel in solids] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Elastic Force (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Groups share their classification results in a class discussion. Teacher builds a shared class results table on the board. Learners then attempt to explain their observations at the particle level using a simple diagram prompt: two circles representing material particles connected by a spring-like bond, and the question 'What happens to the particles when you stretch the material?' Learners draw a before-stretching and after-stretching sketch in their notebooks for an elastic material and for a plastic material, labelling what they think happens to the particle arrangement. Teacher introduces the formal terms elastic behaviour and plastic deformation through a brief, evidence-linked mini-explanation (5 minutes maximum), explicitly connecting each term to the Kericho phenomenon: rope A showed elastic behaviour, rope B showed plastic deformation before it snapped.	Teacher facilitates class sharing: 'Group three, what did you observe with the copper wire after removing the masses?' After groups share, teacher says: 'So we are seeing a pattern — some materials snap back, some stay deformed. Scientists call these two behaviours elastic and plastic. The word elastic does not just mean stretchy — it means the material returns to its original shape and size when the force is removed.' Teacher draws particle diagram on board: 'Imagine the particles inside the sisal rope are held together like beads on a spring. In an elastic material, the springs stretch but pull the beads back. In a plastic material, the beads slide past each other and cannot go back.' WAIT TIME: After drawing the diagram, teacher pauses 30 seconds: 'Look at my diagram. In your notebook, add one thing that is the same between elastic and plastic behaviour, and one thing that is different. You have 30 seconds — go.' Teacher uses cold-call (random name selection) to hear two responses before proceeding.	Particle-level model building bridges observable behaviour to underlying structure. The same-different comparison prompt activates analytical thinking. Anchoring formal vocabulary to the Kericho phenomenon ensures terms are meaningful rather than abstract.	Teacher scans particle diagrams during cold-call responses. Key indicators: Does the learner show that in elastic behaviour the particle spacing returns to normal? Does the plastic diagram show permanent rearrangement? Misconception to address: learners who draw the plastic material as having broken bonds rather than rearranged particles — teacher notes this for Lesson 5 on ductility and brittleness.
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Why S-waves only travel in solids] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Each learner receives two fresh sticky notes. On the first (yellow), they write one question they still have about why the ropes behaved differently — something they do not yet understand. On the second (pink), they write one question about how this knowledge might matter in real life in Kenya. Learners post their notes on a large sheet of paper (or a designated section of the	Teacher introduces the DQB: 'This board is our scientific notebook as a class. Every question you post here is a real scientific question — it deserves a real scientific answer. Over the next six lessons, we are going to come back to this board and answer these questions one by one using evidence.' Teacher reads three or four learner questions aloud with genuine enthusiasm: 'Someone is asking	The DQB externalises learner thinking, creates shared ownership of the investigation, and provides a visible record of intellectual progress across the unit. Grouping questions into themes (material properties, particle behaviour, engineering applications) previews the unit storyline without pre-teaching content.	Teacher photographs or reads all DQB sticky notes after class. Analyse: What proportion of questions are about physical properties vs engineering applications vs particle behaviour? This guides emphasis in subsequent lessons. Note any misconceptions embedded in questions (e.g. 'Does thicker always mean stronger?') — these will be deliberately addressed in

📖 **HTML:** [Elastic Force (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	whiteboard) labelled the Driving Question Board. The class gathers around the DQB and the teacher facilitates a 3-minute look at the questions, grouping similar ones aloud. The teacher posts the unit Driving Question at the top of the board: 'Why do some materials stretch and snap back while others permanently deform or break — and how can we use this knowledge to choose the right material for a safe structure in Kenya?' Learners copy the DQB categories into their notebooks as a reference they will update each lesson.	why the older rope snapped and the newer one did not — that is a brilliant engineering question and it connects to something called the elastic limit, which we will investigate in Lesson 2.' Teacher explicitly validates uncertainty: 'If you do not know the answer yet, that is exactly right — that is why we are here.' Teacher points to the driving question and says: 'This is the question that ties everything together. Keep it in mind as we work.'		Lessons 4 and 6.
Model Building Phase 📹 **VIDEO:** [Why S-waves only travel in solids] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Elastic Force (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	In their notebooks, learners sketch an initial explanatory model for the Kericho phenomenon. The model must include: (1) a simple drawing of rope A and rope B under the 80 kg load, (2) particle-level diagrams for each rope showing what they think is happening inside the material, (3) a written sentence predicting what property of rope B caused it to snap while rope A survived. Learners label their model as 'Initial Model — Lesson 1' because they will revise it in every subsequent lesson. One volunteer shares their model under the document camera or on the board. Class gives one star (something the model explains well) and one question (something the model does not yet explain).	Teacher sets the task: 'Your model does not need to be perfect — it needs to be honest about what you currently think and what you do not yet understand. In Lesson 6 we will look back at this model and see how far your thinking has grown.' Teacher monitors notebook work, looking for integration of particle diagrams with the rope story. During the star-and-question protocol, teacher ensures the question identified by the class is recorded on the DQB: 'That is a new question for our board — let me add it.' Teacher closes the lesson: 'Today you have seen that the same force applied to similar-looking materials can produce very different results. You have started to explain this using particle ideas. In our next lesson, we will measure this behaviour precisely — using a spring, masses, and a ruler — to find out exactly how far a material can be stretched before it stops behaving elastically. Hold your question: where is the boundary between elastic and plastic behaviour?'	Initial model construction commits learners to an evidence-based explanatory position and makes thinking visible for teacher and peer assessment. The star-and-question protocol models scientific peer review and ensures the lesson closes with productive uncertainty rather than false closure.	Collect or photograph initial models. Key rubric: (1) Is elastic behaviour linked to particle spacing returning to normal? (2) Is plastic deformation linked to permanent change in particle arrangement? (3) Is there any attempt to explain why rope B and not rope A? Models showing only surface-level explanations (rope B was old) without particle-level reasoning flag learners who need additional scaffolding in Lessons 2 and 3.

D. TEACHER REFLECTION
After the lesson, consider: Did learners engage genuinely with the Kericho phenomenon or did it feel contrived? Were predictions varied enough to suggest real prior knowledge activation? Did the hands-on exploration generate observable differences between elastic and plastic materials with the available resources? Were learners able to connect particle diagrams to the macroscopic behaviour they observed? Review DQB sticky notes to identify the two or three most common questions — these should be explicitly addressed in Lesson 2 to reinforce that the DQB is a living investigative tool, not a decorative display. If copper wire was difficult to stretch visibly with 200 g, consider replacing it with a longer piece (30 cm) or a softer gauge for future lessons. Ensure initial models are stored safely so learners can access and revise them in Lesson 6.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	One sisal rope stretched and returned to its original shape when the load was removed, while a second sisal rope stretched permanently and then snapped under the same 80 kg load. In the hands-on exploration, rubber bands returned to their original length (elastic behaviour), sisal twine showed slight permanent deformation (borderline plastic), and copper wire remained bent after the load was removed (plastic behaviour).
What did I learn?	Elastic behaviour describes a material that returns to its original shape and size after the applied force is removed. Plastic deformation describes a material that remains permanently changed in shape after the force is removed. These behaviours can be explained at the particle level: in elastic materials, particle bonds stretch and return; in plastic materials, particles shift into new permanent positions. Both the Kericho ropes appeared similar on the outside, but their internal structure and condition determined their behaviour under load.
How does this explain the phenomenon?	The new sisal rope (rope A) behaved elastically because its particle bonds could stretch and recover under the 80 kg load. The old sisal rope (rope B) underwent plastic deformation and eventually fractured because its internal structure had degraded — reducing its ability to recover from stretching. This introduces the key question for the unit: what specific measurable properties determine whether a material will behave elastically or plastically under a given force?

LESSON 2 (80 minutes): Stretching the Truth: Elastic vs. Plastic Behaviour in Real Materials	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to distinguish between elastic and plastic behaviour in materials through hands-on investigation, and to begin building a particle-level model that explains why different materials respond differently to applied forces.
Knowledge	Learners will be able to: (a) define elasticity and plasticity as mechanical properties of materials; (b) identify the elastic limit as the point beyond which a material does not return to its original shape; (c) describe, in qualitative terms, how particle arrangement and bonding relate to elastic and plastic behaviour.
Skills	Learners will be able to: (a) plan and carry out a controlled investigation stretching rubber bands, sisal twine, and copper wire under increasing loads; (b) record, organise, and interpret observational data in a results table; (c) construct an initial particle-level diagram (model) to represent elastic and plastic deformation.
Attitudes	Learners will: (a) appreciate that careful, systematic observation is the foundation of scientific explanation; (b) value collaborative inquiry as a way of building shared understanding; (c) show curiosity and open-mindedness when evidence challenges their initial predictions.
Key Inquiry Question	What determines how a material responds when a force is applied to it?
Purpose in Storyline	In Lesson 1 learners were introduced to the Kericho bridge-rope phenomenon and recorded their initial predictions on the Driving Question Board. In this lesson they gather the first direct evidence: by stretching three different materials (rubber band, sisal twine, copper wire) under controlled loads they observe elastic and plastic behaviour first-hand, giving them the empirical foundation needed to begin explaining WHY the old sisal rope snapped while the new one bounced back.
Safety Notes	⚠️ Safety Notes: (1) Copper wire ends may be sharp — learners must handle cut ends carefully and the teacher should pre-cut wire lengths before the lesson. (2) Hanging masses must be added one at a time; learners must stand to the side, not directly below the hanging load, in case a clamp or retort stand tips. (3) Rubber bands under tension may snap and recoil — learners must wear safety glasses or position a cardboard shield. (4) Sisal twine fibres can irritate skin — learners with sensitive skin should handle twine with gloves. (5) All retort stands must be clamped to the bench edge or weighted at the base.

B. LESSON OVERVIEW

Learners open by revisiting their Lesson 1 predictions on the Driving Question Board (DQB), then individually predict what will happen to three different materials — a rubber band, a piece of sisal twine, and a short length of copper wire — when increasing masses are hung from them. Groups then carry out a structured but open-ended stretching investigation, measuring extension for each material and observing whether the material returns to its original length after the load is removed. Data are recorded in a shared results table. In the Explain phase, the class uses a "claim–evidence–reasoning" (CER) framework to define elasticity and plasticity and to draw a first particle-level model. The lesson closes with a DQB update and a cumulative model-building activity in which each group adds a new panel to their classroom anchor-phenomenon model of the Kericho bridge ropes.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Predict Phase 📹 **VIDEO:** Why S-waves only travel in solids Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Elastic Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners individually examine three labelled samples laid on their desks: (A) a rubber band, (B) a 20 cm length of sisal twine (same fibre type as the Kericho bridge ropes), and (C) a 20 cm length of thin copper wire. Without touching the samples yet, each learner silently records answers to three prediction prompts in their science journal: 1. 'When I hang a 200 g mass from each material, I predict it will stretch by approximately ___ cm.' 2. 'When the mass is removed, I predict the material will / will not return to its original length because _____.' 3. 'Which material do I think behaves most like the NEW sisal rope on the Kericho bridge? Which behaves most like the OLD rope that snapped? Why?' After 4 minutes of silent writing, learners share predictions in their groups of four, identifying where they agree and disagree. Each group nominates ONE point of disagreement to share with the class.	Before distributing samples, say exactly: 'Do NOT touch the materials yet — first, let your brain do the work. You have 4 minutes of silent prediction time. Go.' [Set a visible 4-minute timer.] Circulate silently, reading journals over shoulders. Note 2–3 examples of student reasoning (both correct and misconceived) to reference later — do NOT correct at this stage. After 4 minutes: 'Now share with your group. Your job is to find ONE thing your group disagrees about.' [Allow 3 minutes of group talk.] Cold-call 3 groups to share their disagreement point. Use the sentence stem: 'Group [name], what did your group disagree about?' [WAIT 10 seconds after each response before probing.] Record disagreements on the board under the heading 'What we're not sure about yet.' Do NOT resolve them — say: 'Hold those disagreements. The experiment will give us evidence.'	Strategy: PREDICTION JOURNALING with DISAGREEMENT SURFACING. By requiring individual silent prediction before group talk, this strategy prevents groupthink and ensures every learner commits to a testable claim. Surfacing group disagreements publicly creates cognitive dissonance — a motivational 'need to know' — that drives engagement in the observation phase. It also gives the teacher diagnostic data on prior conceptions (e.g., many learners predict copper wire will stretch the most because it is a metal and 'metals are soft').	Observable indicator: Every learner has written at least one reasoned prediction (not just a guess) in their journal before group discussion begins. Teacher scans journals during circulation. Red flag: learners who write 'I don't know' — prompt them with 'Think about the Kericho rope — did the new rope stretch? Did it come back?' to anchor prediction to prior lesson context.
Observe Phase 📹 **VIDEO:** Why S-waves only travel in solids Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Elastic Force	Groups of four carry out a structured stretching investigation at their benches. Each group receives: – 1 retort stand with clamp and boss – 1 metre ruler (taped vertically to the retort stand) – Samples A, B, C (rubber band, sisal twine, copper wire), each pre-cut to exactly 20 cm and marked at the top	Demonstrate Step 1 and Step 2 with your own retort stand at the front before releasing groups to work. Say: 'I am measuring from the bottom of the clamp to the bottom of the sample — always the same reference point. This is how we control our measurement.' Once groups begin, circulate every 3–4 minutes. At each group, ask ONE specific probe — rotate	Strategy: STRUCTURED DATA COLLECTION with ANOMALY FLAGGING. The ★ marker instruction trains learners to identify the elastic limit empirically — they are discovering the concept before it is named. The 'returned to original?' binary column forces a precise yes/no observational commitment rather than vague qualitative impressions. Photographing results tables creates a data	Observable indicators: (1) Each group's results table has data for all three materials and all five load increments (or a recorded breaking point). (2) At least one row per material is marked with ★ (elastic limit observation) — if no group found it for rubber band, teacher prompts: 'Did it ALWAYS come back perfectly? Look carefully at your last two rows.' (3) Groups correctly calculate extension as (length under load) −

(Flexbook) Source: CK-12 🔍 Search ARES for similar readings	and bottom with a permanent marker – A set of 5 × 100 g slotted masses on a mass hanger – A results table (pre-printed on A3 paper) with columns: Material | Original Length (cm) | Load Added (g) | Length Under Load (cm) | Extension (cm) | Length After Load Removed (cm) | Returned to Original? (Y/N) Procedure (teacher-guided, step by step on the board): Step 1 — Attach sample A (rubber band) to the clamp at the top; measure and record original length. Step 2 — Add 100 g mass hanger. Measure and record length under load. Remove mass. Record length after removal. Step 3 — Repeat, adding one 100 g mass at a time up to 500 g total (or until sample breaks — record this!). Step 4 — Repeat entire procedure for samples B (sisal twine) and C (copper wire). Key observation prompt (printed on the results table): 'At what load — if any — did the material STOP returning to its original length? Mark this row with a ★.' Groups photograph their results table with a tablet/phone for later reference.	through these: – 'What is the extension right now? How did you calculate it?' – 'Has it returned to its original length after you removed the mass? How do you know?' – 'Is there a load where it stopped coming back? Have you marked it?' [WAIT 10–12 seconds after each probe before accepting an answer.] If a group's copper wire permanently bends after 200 g, say: 'Interesting — remember this. Does this remind you of anything about the old Kericho rope?' [Do not answer — plant the connection.] At 20 minutes, give a 2-minute warning: 'Finish your last measurement and begin completing the final column of your table.' After 25 minutes, cold-call one reporter from each of 3 groups to share ONE surprising observation. [WAIT 10 seconds per response.]	artefact learners can return to in later lessons when Hooke's Law is introduced quantitatively.	(original length) — teacher checks arithmetic on two tables during circulation.
Explain Phase 📹 **VIDEO:** Why S-waves only travel in	Whole-class data consolidation and sensemaking using the Claim–Evidence–Reasoning (CER) framework.	During CER writing (Step 2), circulate and place a small green dot sticker on journals where you see strong reasoning to share. Say	Strategy: CLAIM–EVIDENCE–REASONING (CER) FRAMEWORK. CER externalises the structure of scientific	Observable indicators: (1) Each learner's CER statement contains all three components (not just a claim). (2) The 'reasoning' section

solids Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Elastic Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Step 1 (5 min): Teacher posts a class results table on the board (blank template drawn on the chalkboard). Three volunteer reporters fill in their group's ★ rows for each material. Class discusses: 'Did different groups get the same elastic limit for sisal twine?' (Likely not — opens a conversation about measurement variability and the importance of repeating experiments.) Step 2 (8 min): Learners individually write a CER statement in their journals responding to the prompt: 'CLAIM: Material _____ is elastic / plastic because… EVIDENCE: When we added ___ g, the extension was ___ cm and after removing the load the material [did / did not] return. REASONING: This means that elasticity is…' Step 3 (7 min): Teacher leads a whole-class discussion to formalise two definitions on the board: – ELASTICITY: The ability of a material to return to its original shape and size after a deforming force is removed. – PLASTIC BEHAVIOUR (PLASTICITY): Permanent deformation that remains after a deforming force is removed. Teacher then introduces the term ELASTIC LIMIT: 'The maximum force beyond which a material no longer behaves elastically.' Learners copy definitions and annotate them with an example from their own data.	nothing yet. After 8 minutes, cold-call 3 learners whose journals have green dots to read their CER aloud. After each one, say: 'Who agrees with the REASONING part? Thumbs up, thumbs sideways, or thumbs down.' [WAIT 8 seconds.] Count thumbs — if more than 30% show sideways or down, probe: 'What part of the reasoning are you unsure about?' When writing the formal definitions, say: 'Copy this exactly — these are the scientific definitions you will need for your examinations. But more importantly, tell me — which of these two words describes what happened to the OLD Kericho rope? [WAIT 12 seconds.] Cold-call: 'Amina, what do you think, and what is your evidence from today?' Close the Explain phase by asking: 'We have described WHAT happens. Our next big question is WHY — what is happening inside the material at the particle level? We will start that now.'	argument, requiring learners to separate observation (evidence) from interpretation (reasoning). This is the NGSS Science and Engineering Practice of 'Constructing Explanations.' By anchoring the CER to their own data from the stretching investigation, learners experience the definitions of elasticity and plasticity as conclusions they reached — not facts handed to them — increasing conceptual ownership and retention.	uses the word 'return' or 'original shape' — indicating grasp of the elasticity definition. (3) During the thumbs check, if more than 40% show sideways/down, teacher pauses and re-runs the rubber band data row by row as a class before proceeding. Red flag phrase to listen for: 'elastic means it stretches a lot' — this conflates elasticity with extensibility; address immediately with a counter-example: 'The copper wire also stretched — but is it elastic?'

Driving Question Board (DQB) Creation 📹 **VIDEO:** Why S-waves only travel in solids Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Elastic Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class DQB (the large display created in Lesson 1 showing the Kericho bridge photograph and their initial sticky-note predictions). Each group receives two new sticky notes of a DIFFERENT colour from Lesson 1: – YELLOW sticky note: 'One thing I now KNOW from today's evidence that I did not know before.' – BLUE sticky note: 'One NEW question today's evidence has raised for me.' Groups write and post their notes on the DQB under the headings 'EVIDENCE GATHERED' and 'NEW QUESTIONS.' The teacher reads 3–4 blue notes aloud to the class, highlighting questions that will be answered in upcoming lessons (e.g., 'Why does copper wire not come back?' — leads to Lesson 3 on stress-strain; 'Is there a number that tells us how elastic a material is?' — leads to Hooke's Law).	Say: 'Look at your Lesson 1 prediction sticky notes on the DQB. In silence, decide: was your prediction supported or contradicted by today's evidence? You do NOT need to change your old note — science keeps honest records of what we thought BEFORE and AFTER evidence.' [WAIT 20 seconds of silent reflection.] Distribute sticky notes. Allow 3 minutes for writing. Then invite groups to post notes physically on the DQB. Read 3 blue 'new question' notes aloud. For each one, say: 'This is a brilliant scientific question. Here is where it fits in our investigation sequence —' [point to the lesson sequence chart on the wall]. This models metacognitive awareness of where learners are in the learning journey. Do NOT answer the new questions — say: 'We are going to let the evidence answer these for us in the coming lessons.'	Strategy: TWO-COLOUR DQB UPDATE (Know / New Question). The colour-coding creates a visible, cumulative knowledge map that grows across all 6 lessons. Separating 'things I now know' from 'new questions I now have' reinforces that scientific inquiry is iterative — good evidence generates new questions as much as it resolves old ones. This also gives the teacher a real-time diagnostic of class understanding without a formal test.	Observable indicators: (1) Yellow notes contain a specific claim linked to today's data (not vague: 'I learned about elasticity' but specific: 'Rubber band returned to its original length even after 400 g, but sisal twine did not return after 300 g'). (2) Blue notes are genuine questions (not statements) and relate to the phenomenon. Teacher reads all posted blue notes after class to plan differentiated entry points for Lesson 3.
Model Building Phase 📹 **VIDEO:** Why S-waves only travel in solids Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Elastic Force	Each group opens their 'Anchor Phenomenon Model Booklet' — an A3 folded booklet that travels across all 6 lessons, with one double-page spread per lesson. On today's spread (labelled 'Lesson 2: Elastic vs. Plastic'), learners draw a PARTICLE-LEVEL MODEL in two panels: PANEL A — ELASTIC MATERIAL (rubber band under load, then released): Draw particles before load, under load, and after load	Before learners draw, hold up a pre-drawn teacher example (prepared on A3 paper) showing ONLY the 'before load' panel for both materials — particles arranged in a regular pattern with bond lines. Say: 'I have started the model for you. Your job is to complete the UNDER LOAD and AFTER LOAD REMOVED panels. Think about what happens to the distance between particles — and whether they go back.' [Point to each blank panel.]	Strategy: CUMULATIVE PARTICLE-LEVEL MODEL BUILDING. This is the NGSS Science and Engineering Practice of 'Developing and Using Models.' By constructing a particle-level representation — not just a macroscopic description — learners begin to build mechanistic understanding: WHY materials behave the way they do, not just WHAT they do. The booklet format (one spread per lesson) makes the growth of understanding visible	Observable indicators: (1) Both panels are present and show a before/during/after sequence — not just a static diagram. (2) The elastic model shows particles returning to original spacing; the plastic model shows permanently displaced particles. (3) The connecting sentence correctly identifies the old rope as exhibiting plastic behaviour and references particle displacement. Red flag: particles drawn as fixed dots with no bonds shown — prompt: 'What

(Flexbook) Source: CK-12 🔍 Search ARES for similar readings	removed. Use circles to represent particles and lines to represent bonds. Show that bonds STRETCH but particles RETURN to original positions. PANEL B — PLASTIC MATERIAL (copper wire under load, then released): Draw same sequence but show that particles SHIFT to new positions permanently — bonds are disrupted or broken and do not fully restore. Below both panels, each learner writes one sentence connecting the model to the Kericho phenomenon: 'The old sisal rope behaved like Panel _____ because…' Groups share models briefly — teacher selects one 'mentor model' per class to photograph and display.	[WAIT 30 seconds of silent thinking before learners begin drawing.] Circulate. Key probing questions: – 'In your elastic model, why did the particles go back? What pulled them back?' – 'In your plastic model, what happened to the bonds when the wire bent permanently?' – 'Where in your model is the elastic limit — can you show me the exact moment things changed?' [WAIT 10 seconds after each question.] With 3 minutes remaining, say: 'Write your connecting sentence to the Kericho ropes. This is the most important sentence in the whole model — it is WHY we are doing this investigation.' Close with: 'In Lesson 3, we are going to add numbers to this model — we will find out how to measure HOW elastic a material is, not just whether it is elastic or not.'	and tangible across the unit. Connecting each model panel back to the Kericho phenomenon maintains coherence with the anchoring phenomenon throughout.	is holding the particles together? Can you show that connection?' Exit check: teacher collects one model booklet per group at the end of class to review overnight.

D. TEACHER REFLECTION

After this lesson, reflect on the following questions before planning Lesson 3:\n\n1. PREDICTION QUALITY: Did the majority of learners write reasoned predictions (not just guesses)? If predictions were mostly vague ('it will stretch a bit'), consider adding a 'because' sentence-stem scaffold to the prediction prompt in future lessons.\n\n2. DATA QUALITY: Did groups successfully identify the elastic limit (★ row) for at least two of the three materials? If the sisal twine data was inconsistent across groups, this likely reflects measurement technique variability — consider a 5-minute class norming activity at the start of Lesson 3 where one group demonstrates technique before all groups repeat.\n\n3. CER REASONING: Were learners able to write the REASONING component — or did they conflate evidence with reasoning? If more than half the class wrote CER statements with no distinct reasoning, introduce a 'because this means that…' bridge phrase in Lesson 3.\n\n4. DQB BLUE NOTES: What are the top 3 new questions learners posted? Do they reveal misconceptions (e.g., 'does bigger material = more elastic?') or productive lines of inquiry (e.g., 'can you make a material more elastic?')? Plan your Lesson 3 opening hook to address the most common genuine question.\n\n5. PARTICLE MODELS: Did learners draw bonds

(inter-particle forces) or just isolated particles? If bonds were absent, spend 5 minutes at the start of Lesson 3 revisiting particle models from their prior chemistry knowledge before extending into stress-strain concepts.\n\n6. KERICHO CONNECTION: Did learners consistently connect their observations to the phenomenon? If the Kericho bridge was mentioned only in the final model sentence and not during the experiment, build in a 'phenomenon check-in' prompt card on each group's bench in Lesson 3: 'How does what you are observing RIGHT NOW help explain what happened at the Kericho bridge?'

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners stretched rubber bands, sisal twine, and copper wire under increasing loads (100 g to 500 g), recording extension and whether each material returned to its original length after the load was removed. They observed that the rubber band consistently returned to its original length across all loads tested, the sisal twine returned at low loads but showed permanent lengthening beyond approximately 300 g, and the copper wire permanently bent after even a small load of 100–200 g.
What did I learn?	Learners established definitions for elasticity (a material's ability to return to its original shape after a deforming force is removed) and plasticity (permanent deformation after a deforming force is removed). They identified the elastic limit as the critical boundary between these two behaviours, and connected these concepts to the Kericho bridge phenomenon by classifying the new rope as elastic and the old, fatigued rope as having exceeded its elastic limit and behaving plastically before snapping.
How does this explain the phenomenon?	Using a particle-level model, learners began to explain elastic and plastic behaviour in terms of inter-particle bonds: in elastic materials, applied force stretches inter-particle bonds but does not break them, so particles return to their original positions when the force is removed; in plastic materials, the applied force displaces particles beyond the range of their restoring bonds, leaving them in new permanent positions. This model is initial and qualitative — it will be refined and made quantitative in Lessons 3 and 4 when Hooke's Law and stress-strain graphs are introduced.

LESSON 3 (40 minutes): Elastic Potential Energy and the Elastic Limit: How Much Can a Material Store Before It Fails?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners calculate elastic potential energy stored in a stretched spring using Ep = half kx squared, interpret the significance of the elastic limit as a failure threshold, and apply these ideas quantitatively to predict when a material will fail — connecting directly to why the older Kericho sisal rope snapped.
Knowledge	Learners will know: (1) elastic potential energy is the energy stored in a stretched or compressed elastic material, given by Ep = half kx squared; (2) the elastic limit is the maximum extension beyond which a material no longer obeys Hooke's Law and permanent deformation or failure occurs; (3) a safe working load is the maximum load a material can bear while remaining within its elastic limit; (4) the area under a force-extension graph equals the elastic potential energy stored.
Skills	Learners will be able to: calculate Ep using Ep = half kx squared given k and x; read the elastic limit from a force-extension graph; determine the safe working load from a given spring constant; interpret the area under a force-extension graph as energy; apply Hooke's Law quantitatively in Kenyan engineering contexts.
Attitudes	Learners appreciate that quantitative prediction — not guesswork — is the foundation of safe engineering decisions, and develop confidence in using mathematical tools to explain real-world failures such as the Kericho bridge rope snap.
Key Inquiry Question	What determines how a material responds when a force is applied to it? Specifically — how much energy can a material store safely, and what happens when that limit is exceeded?
Purpose in Storyline	Lesson 3 moves learners from qualitative classification (elastic vs. plastic, Lessons 1 and 2) to quantitative explanation. Using force-extension graphs from Lesson 2, learners now calculate the energy stored in the Kericho ropes and identify the precise point — the elastic limit — beyond which the older rope could not recover. This provides the first fully evidence-based, mathematical answer to the driving question and prepares learners for Lesson 4 (stress and strain) by establishing that material failure is predictable and measurable.
Safety Notes	No new practical equipment introduced in this lesson. If learners refer back to Lesson 2 springs, remind them that stretched springs recoil sharply — keep faces away from the line of stretch. All calculations use provided data; no live loading is performed. Learners with visual impairments should be paired with a peer reader for graph-interpretation tasks.

B. LESSON OVERVIEW

Learners open by revisiting the Kericho bridge photograph and their Lesson 2 force-extension graphs. In the Predict phase they estimate — without calculation — how much energy they think was stored in each rope just before the older one snapped, anchoring the lesson in the phenomenon. In the Observe phase the teacher uses a large projected force-extension graph (drawn from Lesson 2 data) and physically shades the triangle under the linear region, making the energy-as-area concept visible before any formula is introduced. In the Explain phase learners derive Ep = half kx squared from the graph area, work through two Kenyan-context problems (vehicle suspension spring on a matatu and a bungee cord at a Nairobi adventure park), and formally define safe working load. In the DQB phase learners add new quantitative questions and revise prior sticky notes. In the Model Building phase learners annotate a printed force-extension graph of the two Kericho ropes, marking the elastic limit, shading the stored energy, and writing a one-paragraph causal explanation of the rope failure — the core evidence artefact for the unit driving question.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Potential energy Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Elastic Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners open their notebooks to their Lesson 2 force-extension graphs. The teacher displays the Kericho bridge photograph and poses the prompt: 'Before we use any formula today, I want you to think about energy. When you stretched your spring in Lesson 2 and then let it go, the spring bounced back. Where did the energy go while the spring was stretched?' Learners write a one-sentence personal prediction in their notebooks, then share with a partner using a Think-Pair-Share. Pairs record one joint prediction on a sticky note: 'We think the energy was stored as ... and when the rope snapped, that energy went to ...' Sticky notes are placed on the DQB under a new column labelled ENERGY QUESTIONS. The teacher then asks: 'Which rope — the new one or the older Kericho rope — do you think stored more energy just before it snapped? Why?' Learners vote by show of hands and justify verbally.	Display the Kericho bridge photograph on the projector. Say: 'Last lesson you measured how much a spring stretched. Today we ask a different question: how much energy was hiding inside that stretched spring? Think quietly for 30 seconds — do not write yet.' WAIT 30 SECONDS. Then say: 'Now write your one-sentence prediction.' WAIT 60 SECONDS for writing. Circulate and read two or three predictions silently — select one that mentions stored energy and one that does not, to use as discussion anchors. After pair-share, cold-call one pair: 'Tell us your prediction and your reasoning.' After the hand vote on which rope stored more energy, say: 'Interesting — we have a split in the room. Hold that idea. By the end of this lesson, your graphs will give you the answer.' Do NOT confirm or correct predictions at this stage. This preserves productive uncertainty.	Think-Pair-Share with sticky-note DQB contribution. Anchoring prediction in the phenomenon keeps energy abstract from becoming disconnected from context. The hand vote creates a visible class disagreement that motivates the Observe phase.	Teacher reads sticky notes to check whether learners associate stored energy with the stretched state of the material (correct) or with the force applied (common misconception). Learners who write 'energy was stored in the hand pulling the spring' rather than 'in the spring itself' are flagged for targeted questioning during the Explain phase.
Observe Phase 📹 **VIDEO:** Potential energy Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Elastic Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher projects a large, clean force-extension graph drawn from the class data of Lesson 2 (a steel spring, linear up to the elastic limit, then curving and dropping). The teacher uses a coloured marker to slowly shade the triangular area under the straight-line region, narrating each step. Learners are asked to observe and describe what the shaded area looks like geometrically. The teacher then draws a vertical dashed line at the elastic limit and shades the area beyond it in a contrasting colour, asking: 'What is different about the shape of the area beyond the	Stand at the projected graph and say: 'Watch carefully — I am not writing a formula yet. I am just colouring.' Shade the triangle slowly. Ask: 'What shape is this?' WAIT 10 SECONDS. Accept geometric descriptions. Ask: 'What does area represent in a graph of force against extension — what are the units?' WAIT 20 SECONDS. Guide learners: 'Force is in newtons. Extension is in metres. Newton times metre equals joule — so the area is energy.' Write AREA = ENERGY on the board in large text. Then say: 'Now I want you to calculate	Concept-first, formula-second sequence: learners observe the area before they see the equation. This prevents the common misconception that Ep = half kx squared is an arbitrary formula rather than a geometric consequence of the force-extension relationship. The Kericho data card keeps the observation anchored to the phenomenon.	Mini-whiteboard or notebook calculation of the triangle area. Teacher scans answers: correct answer for new rope is 0.025 J; correct answer for older rope is 0.18 J. Learners who calculate force times extension without the half factor have not recognised the triangular (not rectangular) area — reteach using the rectangle split into two triangles analogy.

	elastic limit?' Learners observe that it is no longer a neat triangle. The teacher places a transparent ruler along the base and height of the triangle and says: 'This triangle has a base of x — the extension — and a height of F — the force. What formula gives the area of a triangle?' Learners call out: half base times height. The teacher writes half times F times x on the board. Using a printed data card (spring constant k = 40 N per m, extension x = 0.05 m for the new rope; spring constant k = 25 N per m, extension x = 0.12 m for the older rope — simplified values for the Kericho scenario), learners individually calculate the triangle area for each rope on mini-whiteboards or in notebooks before the formula Ep = half kx squared is formally written.	the area of this triangle using half base times height. Use the values on this data card. Do not use a formula I have not written yet — just geometry.' WAIT 90 SECONDS for individual calculation. After learners share answers, say: 'Notice that half times F times x is the same as half times kx times x, because Hooke's Law tells us F equals kx. So half F x equals half kx squared. That is our formula.' Write Ep = half kx squared. Emphasise: 'The formula came from your graph — it is not magic, it is geometry.'		
Explain Phase 📹 **VIDEO:** Potential energy Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Elastic Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners now work through three structured tasks in their notebooks. Task 1 — Deriving the formula: Learners write out the steps from half Fx to half kx squared substituting Hooke's Law, confirming the formula in their own handwriting. Task 2 — Kenyan worked example A (Vehicle suspension spring): A matatu in Nairobi has a suspension spring with k = 18 000 N per m. When fully loaded with 8 passengers (total additional load of 5400 N), the spring compresses by 0.30 m. Learners calculate: (a) the elastic potential energy stored in the spring at full load using Ep = half kx squared; (b) whether this load exceeds the elastic limit given that the spring's elastic limit is reached at an extension of 0.35 m. They conclude whether the spring is operating safely. Task 3 —	Distribute the worked-example worksheet (or write problems on the board). Say: 'Work individually and silently for 8 minutes. Show every step — I am checking your substitution, not just your answer.' WAIT 8 MINUTES, circulating and noting common errors (unit errors: cm instead of m for extension; forgetting to square x). After 8 minutes say: 'Compare your answer for Task 2b with your neighbour. If you disagree, work out where your answers diverge.' WAIT 2 MINUTES. Then cold-call one group for each task answer. For Task 2b ask: 'The extension is 0.30 m and the elastic limit is at 0.35 m — so what is the conclusion? Be precise.' WAIT 15 SECONDS. Accept: 'The spring is operating within its elastic limit — it is safe.' Then introduce safe working load: 'Here is the key	Structured worked examples move from guided (teacher narrates Task 1) to collaborative (Tasks 2 and 3 peer-compared) to independent application. Kenyan contexts (matatu suspension, Nairobi bungee) sustain relevance. The connective question at the end requires learners to synthesise the new quantitative tool with the anchoring phenomenon — this is the key sensemaking moment of the lesson.	Circulate during the 8-minute silent work period. Check: (1) correct substitution into Ep = half kx squared; (2) correct unit for extension (metres); (3) correct comparison of extension to elastic limit for the safe-load judgment. Exit-level check: ask two learners to explain in words what safe working load means without looking at their notebooks.

	Kenyan worked example B (Bungee cord): A bungee cord at a Nairobi adventure park has k = 120 N per m. A participant of mass 65 kg falls and extends the cord by 8.0 m. Calculate the elastic potential energy stored at maximum extension. Learners then discuss: 'If the elastic limit of this cord is reached at 9.5 m, is the participant safe?' After individual work (8 minutes), groups compare answers and the teacher leads a whole-class debrief. The teacher formally introduces safe working load: 'The safe working load is the maximum force we can apply and still stay within the elastic limit — the region where Hooke's Law holds and the material will recover.' Learners write a formal definition in their notebooks and connect it explicitly to the Kericho rope: 'The older rope had a lower elastic limit — its safe working load was less than the weight of the tea sack, which is why it failed.'	engineering term. Say it with me: safe working load.' After the debrief, pose the connective question: 'So — the older Kericho rope snapped. Using what you have just calculated, what does that tell you about the load on the bridge compared to the rope s safe working load?' WAIT 20 SECONDS. Expect learners to state that the load exceeded the safe working load, pushing the rope beyond its elastic limit. Affirm: 'Exactly. The rope was asked to do more than it was built to do — and now we can say that in numbers, not just words.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Potential energy Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Elastic Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the DQB. Each learner receives two sticky notes of a new colour (designated for Lesson 3 contributions). They write: (1) one piece of new evidence they now have that helps answer the driving question — e.g., 'The older rope stored more energy (0.18 J) before snapping because it stretched further past its elastic limit'; (2) one new question the lesson has raised — e.g., 'Does the thickness of the rope affect how much energy it can store before snapping?' Learners post both notes. The class reads the new evidence notes aloud in a gallery walk (2 minutes). The teacher identifies two or three questions that preview	Say: 'The DQB is our living record of this investigation. Today you have new evidence — quantitative evidence. Add it now.' Distribute sticky notes of the new colour. WAIT 2 MINUTES for writing. After the gallery walk, read aloud the questions about cross-sectional area and material thickness. Say: 'I am circling these. Notice that you asked almost exactly the same question that engineers in Kenya ask when they design bridges. That question has a name — tensile stress — and it is the focus of Lesson 4.' Do not answer the question now; preserve curiosity. Ask: 'Has anyone changed their mind about which rope stored more energy — do your answers	The two-sticky-note protocol (evidence plus question) ensures that DQB contributions are both backward-looking (consolidating learning) and forward-looking (generating inquiry momentum). Circling questions that bridge to Lesson 4 makes the storyline progression explicit and gives learners a sense of authorship over the curriculum direction.	Read the evidence sticky notes to check whether learners are citing specific values (e.g., calculated Ep in joules) or remaining vague (e.g., 'the rope had too much energy'). Vague notes indicate that the formula was used procedurally without conceptual connection — these learners are targeted for additional questioning in Lesson 4.

	Lesson 4 content (stress and strain) and circles them with a marker, saying: 'These questions about thickness and cross-section are exactly what Lesson 4 will investigate — you are already thinking like materials scientists.'	match your prediction from the start of the lesson?' WAIT 15 SECONDS for show of hands and brief justification.		
Model Building Phase 📹 **VIDEO:** [Potential energy](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Elastic Force (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Learners receive a printed A4 sheet containing two overlapping force-extension curves: Curve A (new sisal rope — steeper gradient, elastic limit clearly marked at a moderate extension) and Curve B (older sisal rope — shallower gradient, elastic limit marked at a lower extension, with a snap indicated by a cross). Axes are labelled but numerical values are left blank for learners to fill in using reasonable estimates from the lesson context. Learners complete the following model-building tasks in their notebooks: (1) Label the elastic limit on each curve. (2) Shade the elastic potential energy stored in each rope at its elastic limit (the triangle under each curve up to that point). (3) Calculate approximate Ep for each rope using the triangle area (no formula required — geometry only). (4) Write a one-paragraph causal explanation (minimum 4 sentences) answering: 'Why did the older Kericho rope snap while the new rope survived the same load?' The explanation must include the terms: elastic limit, safe working load, elastic potential energy, and Hooke s Law. Groups share paragraphs and offer one piece of peer feedback using the sentence stem: 'Your explanation would be stronger if you included ...' The teacher collects one paragraph per group as the formative evidence artefact for this	Distribute the printed graph sheet. Say: 'This is the most important task of the lesson. You have all the tools — the formula, the graph, the Kericho story. Now I want you to put them together in writing.' WAIT 10 MINUTES for individual work on tasks 1 to 4. Then say: 'Share your paragraph with your group. Give feedback using the sentence stem on the board.' WAIT 3 MINUTES for peer feedback. Then cold-call one group to read their paragraph aloud. After they read, ask the class: 'Did they use all four required terms? Thumbs up or thumbs down.' Address any missing terms. Collect one paragraph per group. Close the lesson by returning to the driving question: 'We started by asking why one rope snapped and one survived. Today you can answer that with a number — the older rope s safe working load was exceeded, the elastic potential energy stored in it at the moment of failure was larger than the material could return elastically, and it crossed the elastic limit. That is a scientific explanation, not a guess.' Say: 'In Lesson 4 we will make this even more precise by looking at how the thickness of the rope matters — and that will bring us to a new quantity called tensile stress.'	The annotated graph serves as a boundary object — it simultaneously represents the mathematical content (area as energy, gradient as k) and the phenomenon (the two Kericho ropes). The written paragraph requires learners to integrate vocabulary, calculation, and narrative — the highest level of sensemaking in this lesson. Peer feedback using a structured stem maintains academic language norms without marginalising less confident writers.	Collected paragraphs are assessed against three criteria: (1) correct use of all four required terms; (2) causal logic (older rope exceeded elastic limit therefore permanent deformation and failure); (3) quantitative reference (some mention of the energy or load values calculated during the lesson). Paragraphs scoring low on criterion 2 (listing facts without causal connectives) indicate that the concept of elastic limit as a failure threshold has not been fully internalised — these learners receive a targeted prompt at the start of Lesson 4.

	lesson.			

D. TEACHER REFLECTION
After the lesson, consider: Did learners arrive at Ep = half kx squared through the geometric argument (area of triangle) or did they simply accept the formula? If the majority copied the formula without engaging with the derivation, repeat the shading-and-geometry step at the start of Lesson 4 using a different spring dataset. Were the Kenyan contexts (matatu, bungee cord) sufficiently motivating, or did learners treat them as abstract word problems? If the latter, consider sourcing a photograph of a Nairobi bungee operation or a matatu suspension spring to display during Task 2 and 3. Review collected paragraphs before Lesson 4: identify whether learners understand the elastic limit as a material property (intrinsic to the rope) rather than a property of the load (extrinsic). This distinction is foundational for the stress and strain concepts in Lesson 4. Note any learners who conflated elastic potential energy with gravitational potential energy — this cross-topic confusion will resurface when learners calculate energy in the bungee scenario and should be addressed explicitly in the Lesson 4 opener.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Force-extension graphs from Lesson 2 showing the linear (Hooke s Law) region and the point of elastic limit for the steel spring; the triangular area under the force-extension curve shaded by the teacher; the two overlapping force-extension curves representing the new and older Kericho sisal ropes, with different gradients and different elastic limits marked.
What did I learn?	Elastic potential energy is stored in a stretched material and equals half kx squared, which is equivalent to the area of the triangle under the force-extension graph in the linear region; the elastic limit is the boundary beyond which a material cannot return to its original shape and may fail permanently; the safe working load is the maximum load a material can bear while remaining within its elastic limit; the older Kericho rope failed because the load exceeded its safe working load, pushing it past its elastic limit and causing permanent deformation and snap.
How does this explain the phenomenon?	The older Kericho sisal rope snapped because the weight of the tea sack (approximately 800 N) exceeded the rope s safe working load — that is, the force corresponding to the rope s elastic limit. Beyond that limit, the material could no longer store and return energy elastically; instead it deformed plastically and ultimately fractured. The new rope, with a higher elastic limit and greater spring constant, remained within its safe working load under the same force and so recovered elastically when the load was removed. This explanation is now supported by quantitative calculation of elastic potential energy and direct reading of the elastic limit from the force-extension graph.

LESSON 4 (80 minutes): Hooke's Law — Measuring Spring Constant and Finding the Elastic Limit	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to investigate the relationship between force and extension in a spring, determine the spring constant, and identify the elastic limit — connecting quantitative measurements to the behaviour of the Kericho bridge ropes.
Knowledge	By the end of this lesson, learners should be able to: (a) state Hooke's Law and write the mathematical relationship $F = kx$; (b) define the spring constant k and the elastic limit; (c) interpret a force-extension graph to determine k from the gradient and identify the elastic limit.
Skills	By the end of this lesson, learners should be able to: (a) conduct a controlled investigation measuring spring extensions for increasing loads; (b) record data accurately in a results table; (c) plot a force-extension graph and calculate the gradient; (d) compare k values across different springs and make evidence-based conclusions.
Attitudes	Learners appreciate that precise, quantitative measurement — rather than visual inspection alone — is the only reliable way to predict material behaviour and prevent structural failures such as the snapping of the Kericho bridge rope.
Key Inquiry Question	What determines how a material responds when a force is applied to it?
Purpose in Storyline	Lessons 1–3 established qualitative differences between elastic and plastic behaviour and introduced stress and strain conceptually. Lesson 4 is the quantitative pivot: learners now put numbers to elastic behaviour by determining spring constants and pinpointing the elastic limit on a graph. This directly answers the DQB question — 'Can we predict which rope will snap?' — by showing that every spring (and every rope) has a measurable limit beyond which permanent deformation begins. Lesson 5 will extend this to tensile strength and breaking point.
Safety Notes	1. Slotted masses must be loaded and removed gently — sudden release can cause the retort stand to topple. Clamp the retort stand to the bench or place a sandbag at the base. 2. Do NOT load springs beyond 5 N during this lesson; exceeding the design limit may cause the spring to fly off and injure eyes — learners must wear safety goggles. 3. If a spring snaps, call the teacher immediately; do not handle broken wire ends. 4. Keep the area under hanging masses clear of feet and bags at all times.

B. LESSON OVERVIEW
Learners conduct a structured, quantitative investigation of Hooke's Law using a steel spring suspended from a retort stand. Working in groups of four, they add slotted masses (0.1 kg increments, up to 0.5 kg / 5 N), record the extension of the spring at each load, and plot a force-extension graph on graph paper. From the linear portion of the graph they calculate the gradient to find the spring constant k (N/m). They then identify the elastic limit — the point at which the graph curves away from linearity — and compare k values between a stiff spring and a soft spring. The lesson closes with a structured DQB update: learners propose that the older Kericho rope had a lower effective k and a lower elastic limit, making it more likely to reach permanent deformation under the 80 kg tea-leaf sack.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Each learner receives a half-sheet 'Prediction Slip' with two diagrams:	Distribute prediction slips face-down; say 'Do NOT flip until I say	Prediction Slip with Sketch Graph — activates prior knowledge of	Teacher scans prediction slips during pair-share for two

📹 **VIDEO:** [Video: Hooke's Law] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Newton's Law of Gravity (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	(A) a spring with a 1 N load showing a 2 cm extension, and (B) the same spring with a 4 N load — the extension box is blank. Learners individually write their predicted extension for (B) and a one-sentence justification. They then share with a shoulder partner for 90 seconds before the teacher facilitates a whole-class share-out. A second prompt on the slip asks: 'If you kept adding more and more mass to the spring, what do you think would eventually happen — and how would you see this on a graph?' Learners sketch a rough prediction graph on the back of the slip. These prediction slips are kept and revisited during the Explain phase.	go.' After 30 seconds of silent think time: 'Flip your slips — you have 3 minutes to write your prediction and justification individually. No talking yet.' After 3 minutes: 'Turn and share with your shoulder partner — you have 90 seconds.' Cold-call 3 pairs (choose one high-confidence and one uncertain pair): 'Amina, what extension did your pair predict for 4 N — and why?' After each response, ask the class: 'Hands up if your prediction matches — hands up if yours is different.' Do NOT confirm or deny. Say: 'Hold that thought — the spring on the retort stand will answer this for us in a few minutes.' Pin 3–4 varied prediction graphs on the board under the label 'OUR PREDICTIONS — CHECK THESE AT THE END.'	proportionality from mathematics (Grade 9 linear relationships) and forces learners to commit to a quantitative claim before gathering data, creating cognitive tension that motivates careful measurement. Shoulder-pair sharing surfaces misconceptions (e.g., 'it doubles every time' vs. 'it eventually stops stretching') before instruction.	indicators: (1) Does the learner predict a proportional doubling (8 cm) — indicating readiness for Hooke's Law? (2) Does the learner's sketch graph show a straight line, a curve, or a plateau — revealing their mental model of elastic limits? Teacher notes any learner who predicts 'the spring will not stretch more because it is full' as a priority misconception to address in Phase 3.
Observe Phase 📹 **VIDEO:** [Video: Hooke's Law] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Newton's Law of Gravity (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Groups of four set up a retort stand with a steel spring hanging from the clamp. Each group has: a metre rule (clamped vertically beside the spring), a set of slotted masses (50 g each, up to 500 g), a results table in their science journals, and safety goggles. The investigation procedure is displayed on the board: (1) Record the natural length of the spring (no load) — this is the reference position. (2) Add masses one at a time (0.5 N increments). After each addition, wait 10 seconds for the spring to settle, then record the new length and calculate extension (extension = new length − natural length). (3) Continue until the graph begins to show a curve OR until 5 N is reached — whichever comes first. (4) Record observations in the two-column results table: Force (N) | Extension (cm). Groups using a 'stiff' spring	Before groups begin, demonstrate the setup at the front bench (2 minutes): 'Watch how I read the ruler — my eye must be level with the bottom of the spring coil to avoid parallax error.' Ask: 'Korir, what error am I avoiding by keeping my eye level? Give me 10 seconds.' [WAIT 10 seconds.] Cold-call Korir. Circulate during data collection — every 4 minutes visit a new group. At each group ask: 'What pattern are you noticing in your extension column?' and 'How confident are you in your last reading — did you wait for the spring to stop oscillating?' At the 20-minute mark, prompt groups who have not yet seen a curve: 'Add one more 50 g mass — watch very carefully what happens to the spring after you remove the load.' For groups using the soft spring that reaches its elastic limit early: 'Mark that point with a star in your	Structured Data Collection with Comparative Spring Groups — by deliberately giving half the class a stiff spring and half a soft spring, the investigation produces two different gradients and two different elastic limits. This built-in comparison mirrors the two Kericho bridge ropes and prevents learners from treating Hooke's Law as a single fixed number. The real-time graph plotting (rather than post-hoc) keeps learners engaged with the emerging shape of the relationship.	Teacher checks each group's results table for two indicators: (1) Are extension values increasing by approximately equal amounts for equal force increments (proportionality)? A group whose values jump erratically likely has a parallax reading error — teacher re-demonstrates. (2) Has the group correctly calculated extension as (new length − natural length), not just recorded the new length? Any group recording total length instead of extension is flagged and corrected before graph plotting begins.

	(k ≈ 25 N/m) and groups using a 'soft' spring (k ≈ 10 N/m) are alternated around the room. After collecting data, each group plots their force-extension graph on pre-printed graph axes (Force on y-axis, Extension on x-axis), draws the line of best fit through the linear region, and marks with a cross (×) where the line begins to curve.	table — that is the moment we have been waiting for.' After all groups have plotted their graphs, say: 'Circle the region on your graph where the line is straight — that region has a name we are about to give it.'		
Explain Phase 📹 **VIDEO:** [Video: Hooke's Law] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Newton's Law of Gravity (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The teacher leads a whole-class data collation: one representative from each group writes their spring type and k value (calculated from graph gradient) on the class data wall — a large sheet of paper at the front divided into 'Stiff Spring' and 'Soft Spring' columns. Learners compare values within each column (expecting consistency) and between columns (expecting a clear difference). The teacher then introduces the formal vocabulary: Hooke's Law (F = kx), spring constant k (unit: N/m), elastic limit, and proportionality limit. Learners annotate their graphs with these labels. A worked example is solved together: 'A spring has k = 20 N/m. A Kenyan farmer hangs a 2.5 kg bunch of bananas from it. What is the extension?' (Answer: F = 2.5 × 10 = 25 N; x = F/k = 25/20 = 1.25 m — teacher pauses here to ask 'Does 1.25 m make sense for a small spring?', prompting learners to recognise the spring in the example must be very soft or the bananas are very heavy — reinforcing unit and magnitude sense-checking). Learners then retrieve their prediction slips from Phase 1, compare their predicted graph shape with their actual graph, and write a two-sentence 'I	At the class data wall, ask: 'Wanjiru, read out your group's k value for the stiff spring.' After three groups have shared: 'What do you notice about these numbers? [WAIT 12 seconds.] Fatuma, what pattern do you see?' After the pattern is confirmed: 'Now look at the soft spring column — Otieno, how does this group of k values compare? [WAIT 10 seconds.]' Introduce the formal definition: write F = kx on the board and say: 'k is the spring constant — the larger k is, the stiffer the spring, and the more force you need for every extra centimetre of stretch.' Connect explicitly: 'Now think about our two Kericho ropes — if rope A is like our stiff spring and rope B is like our soft spring, which rope do you think had the higher k? [WAIT 15 seconds.] Cold-call 2 learners, require justification: 'Because…' During the prediction slip review, choose one learner whose original prediction was wrong and celebrate: 'This is exactly what science is — you made a brave prediction, the evidence corrected you, and now you know something true about the world.'	Class Data Wall with Cross-Group Comparison — aggregating data from all groups into a shared display models the scientific practice of using multiple trials and replications to build confidence in a result. The 'I thought… but I found…' reflection is a metacognitive sensemaking tool that helps learners explicitly track conceptual change, a core NGSS Science and Engineering Practice (SEP 6: Constructing Explanations). The F/x ratio check reinforces that k is not read off the graph but is a property derivable from any point in the linear region.	Exit check (2 minutes at end of phase): Teacher shows a force-extension graph on the board with the elastic limit marked and asks three rapid cold-call questions: (1) 'What does the gradient of the straight-line portion represent?' (2) 'What happens to the spring if you load it beyond this marked point?' (3) 'Which spring — stiff or soft — would you choose for a bridge rope, and why?' Observable indicator of success: learner correctly identifies k as gradient, describes permanent deformation beyond the elastic limit, and justifies their rope choice using the language of Hooke's Law.

	thought… but I found…' reflection in their journals. Finally, each group calculates the ratio F/x at three points in the linear region and confirms it is constant — this is the physical meaning of k.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Video: Hooke's Law] Source: PhET Interactive Simulations (English) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Newton's Law of Gravity (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	Each group receives two colours of sticky notes: yellow for 'new understanding' and pink for 'new question.' Groups spend 4 minutes writing: at least one yellow note (e.g., 'The stiff spring has a larger k — it needs more force per centimetre of stretch') and at least one pink note (e.g., 'Does a rope have a spring constant? How do we measure k for a rope made of sisal?' or 'What happens after the elastic limit — does the material keep stretching forever or does it snap?'). Groups post their notes on the class DQB under the existing columns: 'What we think we know' (yellow) and 'New questions' (pink). The teacher reads three pink notes aloud and says: 'These questions are exactly what Lesson 5 will address — tensile strength and the breaking point.' The teacher also draws a large arrow on the DQB connecting today's finding ('k is measurable and different for different materials') to the original phenomenon question ('Can we predict which rope will snap?') and writes: 'PARTIAL ANSWER: A rope with a lower k and a lower elastic limit is more likely to permanently deform and snap under the same load.'	Before sticky-note writing, re-read the driving question aloud: 'Why do some materials stretch and snap back while others permanently deform or break — and how can we use this knowledge to choose the right material for a safe structure in Kenya?' Say: 'Your yellow note must connect something from today's investigation directly to this question.' After notes are posted, select one yellow and one pink note from different groups. Ask the class: 'Ngeno's group wrote that the elastic limit is the key safety boundary. Do you agree? [WAIT 12 seconds.] Baraka, respond to Ngeno's idea.' Facilitate a 2-minute class discussion. Close by pointing to the DQB: 'Notice how our board is growing — two lessons ago we could only say the ropes looked the same but behaved differently. Now we can say: the rope that snapped had passed its elastic limit. Next lesson we find out exactly when a material breaks.'	DQB Sticky-Note Protocol (Two-Colour) — the deliberate colour-coding of 'understanding' versus 'question' notes trains learners in the scientific practice of distinguishing claims from open problems (NGSS SEP 1: Asking Questions and Defining Problems). The teacher's visible arrow connecting today's evidence to the phenomenon creates a public record of the storyline's progress, making the cumulative nature of scientific knowledge construction explicit.	Teacher reads all posted yellow notes after class. Success indicator: yellow notes reference k, elastic limit, or the comparative stiffness of the two spring types in relation to the bridge ropes. A yellow note that says only 'springs stretch when you pull them' signals insufficient conceptual advancement — teacher follows up individually at the start of Lesson 5. Pink notes are catalogued to inform the framing of Lesson 5.
Model Building Phase 📹 **VIDEO:** [Video: Hooke's	Each learner opens to their running 'Rope Model' page in their science journal — a diagram of the two Kericho bridge ropes that has been updated each lesson.	Circulate during model revision. Look for two common errors: (1) Learners who draw both ropes with the same gradient — prompt: 'You said stiff and soft springs	Cumulative Annotated Diagram with Force-Extension Graph Overlay — the running rope model acts as a 'knowledge artefact' that makes learning visible across the	Collect 4–5 science journals (rotating sample each lesson) and check the model revision against three criteria: (1) Rope B graph has a shallower gradient than

Law Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Newton's Law of Gravity (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Today's task is a two-part model revision: (1) Annotate each rope with a spring-constant label — Rope A (survived): 'HIGH k — large restoring force per unit extension' and Rope B (snapped): 'LOW k AND low elastic limit — passed its proportionality limit under the load of the 80 kg tea sack.' (2) On the same page, sketch a force-extension graph for each rope side by side, showing Rope A with a steeper gradient and a higher elastic limit than Rope B. Learners add a vertical dashed line at F = 80 kg × 10 m/s² = 800 N to both graphs and shade the region beyond the elastic limit in red, labelling it 'DANGER ZONE — permanent deformation.' Groups share their revised models with an adjacent group and give one piece of feedback using the sentence frame: 'Your model shows ______ clearly, and you could improve it by ______.'	have different k values. Should your two rope graphs look identical?' (2) Learners who place the 800 N line below the elastic limit for Rope B — prompt: 'The rope DID snap. What does that tell you about where 800 N falls relative to Rope B's elastic limit?' After peer feedback: cold-call one learner to describe their revised model to the class in 30 seconds. Close the lesson: 'Today you turned a qualitative story about two ropes into a quantitative model with real numbers. A structural engineer in Nairobi does exactly this before choosing cables for a bridge. Next lesson, we find the final piece: what force actually breaks a material?'	six-lesson sequence (NGSS SEP 2: Developing and Using Models). By overlaying the F = 800 N line onto their sketched graphs, learners practise using a quantitative model to make a predictive explanation — not just describe what happened, but explain why it had to happen given the material's properties.	Rope A (correct k relationship). (2) Rope B elastic limit is below 800 N (explains why it snapped). (3) Rope A elastic limit is above 800 N (explains why it survived). Any journal missing criterion (2) or (3) indicates the learner has not yet connected Hooke's Law quantitatively to the phenomenon — teacher provides targeted oral feedback at lesson start next day.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your journal:

1. PROPORTIONALITY: Did learners correctly distinguish between the extension (change in length) and the total length of the spring during data collection? If parallax errors were common, consider modelling the correct eye-level reading technique more explicitly at the start of Lesson 5.
2. GRAPH INTERPRETATION: Were learners able to identify the elastic limit as the point where the graph departs from linearity — or did they simply mark the last data point? If the latter, prepare an annotated sample graph for Lesson 5's warm-up.
3. CROSS-GROUP COMPARISON: Did the class data wall reveal consistent k values within each spring type? Large spread within one group may indicate that some groups used worn springs with inconsistent behaviour — replace equipment before Lesson 5.
4. DQB QUALITY: Did the pink 'new question' notes move beyond surface questions (e.g., 'what is the unit of k?') toward deeper mechanistic ones (e.g., 'what is happening to the rope fibres at the microscopic level past the elastic limit?')? If not, model a higher-order question yourself and post it on the DQB as a 'teacher question' to raise the bar.
5. PHENOMENON CONNECTION: Did learners spontaneously use the Kericho rope context in their explanations, or did they treat the spring investigation as an isolated physics activity? If the latter, make the phenomenon connection more explicit at the start of each phase in future lessons — consider posting a photograph of the bridge ropes at the front of the room for the duration of the unit.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners added slotted masses (0–5 N) to a hanging steel spring and measured the extension at each load increment using a metre rule clamped beside the retort stand. Half the class used a stiff spring; half used a soft spring. Learners recorded data in a force-extension table, plotted their results on graph paper, drew a line of best fit through the linear region, and marked the point where the graph began to curve (the elastic limit). They compared k values across groups using a class data wall.
What did I learn?	Learners learned that: (1) For a spring loaded within its elastic limit, extension is directly proportional to the applied force — this is Hooke's Law ($F = kx$). (2) The spring constant k (N/m) is the gradient of the linear portion of the force-extension graph; a larger k means a stiffer spring requiring more force per unit extension. (3) Beyond the elastic limit, the spring no longer returns to its original length when the load is removed — permanent deformation begins. (4) Different springs have different k values and different elastic limits, meaning their behaviour under the same force is quantitatively predictable.
How does this explain the phenomenon?	Learners explained that the Kericho bridge rope that snapped had a lower spring constant and a lower elastic limit than the rope that survived. When the 80 kg tea-leaf sack (≈ 800 N) was placed on the bridge, the force applied to the older rope exceeded its elastic limit, causing permanent deformation in the rope fibres. Because the force continued beyond the elastic limit toward the breaking point, the rope snapped. The rope that survived had a sufficiently high elastic limit that 800 N fell within its proportional region, allowing it to return to its original length when the load was removed. Learners revised their running rope models to show two force-extension graphs side by side, with a vertical line at 800 N demonstrating that Rope B's elastic limit was exceeded while Rope A's was not.

LESSON 5 (80 minutes): Stress, Strain, and the Young's Modulus: Predicting Which Rope Will Snap	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To develop learners' ability to quantify how materials respond to applied forces using the concepts of stress, strain, and Young's Modulus, and to use these measurements to predict material failure — directly linking to the Kericho bridge rope phenomenon.
Knowledge	Learners will define stress ($\sigma = F/A$) and strain ($\varepsilon = \Delta L/L_0$), calculate Young's Modulus ($E = \sigma/\varepsilon$), interpret a stress-strain graph including the elastic limit, yield point, and breaking point, and explain why two ropes of similar appearance can have vastly different Young's Modulus values and failure behaviours.
Skills	Learners will: (a) measure force, cross-sectional area, original length, and extension of a wire/elastic sample; (b) calculate stress, strain, and Young's Modulus from experimental data; (c) plot and interpret a stress-strain graph; (d) use E-values to predict which of two materials will fail first under a given load.
Attitudes	Learners appreciate that precise quantitative measurement — not just appearance — is the foundation of safe engineering decisions, and that Kenyan engineers, bridge builders, and farmers all rely on these principles when selecting materials.
Key Inquiry Question	How do we measure and compare the stiffness of different materials, and how can those measurements tell us in advance which material will deform permanently or snap under a given load?
Purpose in Storyline	In Lessons 1–4, learners identified elastic vs. plastic behaviour, explored Hooke's Law, and investigated elastic potential energy and the elastic limit. They now know that the old Kericho rope crossed its elastic limit. But the class has not yet been able to say WHY two ropes of similar fibre can have such different stiffness and breaking loads. Lesson 5 introduces the quantitative language of stress, strain, and Young's Modulus — giving learners the mathematical tools to compare materials, read engineering data sheets, and finally begin to answer the driving question with numbers, not just words.
Safety Notes	When loading wires or rubber samples to breaking point: (1) All learners must wear safety goggles — a snapping wire can recoil at high speed. (2) Place a folded cloth or cardboard beneath hanging masses to catch falling weights if the sample breaks. (3) No learner should stand directly below a hanging load. (4) Use G-clamps to secure the retort stand to the bench. (5) Masses must be added gently and one at a time — no dropping. Teacher must inspect all clamp connections before the experiment begins.

B. LESSON OVERVIEW

Learners return to the Kericho bridge rope puzzle armed with four lessons of prior knowledge (elastic/plastic behaviour, Hooke's Law, elastic potential energy, elastic limit). Today they discover that visual similarity between ropes is meaningless — what matters are quantitative material properties: stress, strain, and Young's Modulus. Through a structured experiment using rubber strips and thin wire (or sisal strands), learners generate their own stress-strain data, plot graphs, identify key graph features (elastic limit, yield point, UTS, fracture point), and calculate E-values. They then compare their experimental E-values with reference values for sisal (new) and sisal (aged/degraded) to explain the bridge failure in precise scientific language. By the end of the lesson, learners can predict — before a test — which material will snap under the farmer's 80 kg load.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Complex numbers with the same modulus (absolute value) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Sine Graph and Cosine Graph: Changing Amplitude (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown two identical-looking strips of material side by side on the demonstration bench — one fresh rubber band (representing the new sisal rope) and one that has been left in the sun and stretched repeatedly over several days (representing the old, degraded sisal rope). Both are 20 cm long and appear the same width. Each learner silently writes a prediction on a sticky note responding to the prompt: 'I think Strip A / Strip B will snap first because…' and 'I think the strip that does NOT snap will carry the load more safely because…'. After 3 minutes of silent writing, learners share predictions in Think-Pair-Share: first 60 seconds with a partner, then 4 cold-call pairs share with the class. Teacher records the class's predicted reasons on the Driving Question Board under the column 'What we THINK we know — Lesson 5'.	Display both strips on a retort stand with labels 'Rope A — New Sisal' and 'Rope B — Old Sisal (Kericho bridge)'. Say verbatim: 'Before we touch anything, I want your brain's best guess — written down, no talking for three minutes. Remember: the farmer's life may depend on choosing the right rope, so be specific.' Set a visible 3-minute timer. After timer: 'Turn to your partner and share your prediction and your reason. You have 60 seconds. Go.' After 60 seconds, cold-call exactly 4 pairs using the class register (rows 2, 5, 8, 11). For each pair ask: 'What was your prediction AND what reasoning did your group use?' Apply 12-second wait time after each cold-call before accepting the answer. Record key reasoning words verbatim on the DQB (e.g., 'older rope is weaker', 'sun damages fibres'). Do NOT correct predictions — say: 'Interesting. We will come back to see whose prediction the data supports.'	Prediction-before-observation (Predict-Observe-Explain anchor): By committing predictions to writing before any data is gathered, learners activate prior schema (Hooke's Law, elastic limit from Lessons 1–4) and create cognitive dissonance that will be resolved during the experiment. Recording predictions on the DQB makes prior thinking visible and revisable.	Observable indicators: (1) Every learner has a written sticky note — teacher does a 30-second visual scan of the room. (2) Cold-called pairs use lesson-specific vocabulary (elastic, elastic limit, force, deformation) — teacher tallies correct vocabulary use. (3) Teacher notes whether learners reference only appearance ('they look the same') vs. internal material properties — this gap drives the lesson.
Observe Phase 📹 **VIDEO:** Complex numbers with the same modulus (absolute value) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Sine Graph and Cosine Graph: Changing Amplitude (PLIX)	Groups of 4 conduct a structured experiment titled 'The Stress-Strain Investigation'. Each group receives: a rubber strip (Strip A) and a pre-fatigued rubber strip (Strip B), a 30 cm ruler, a vernier calliper (or mm ruler for cross-section), a set of 10 g masses up to 200 g on a mass hanger, a retort stand with clamp, and a data table printed on A4 paper. Step 1 (5 min): Learners measure and record the original length (L_0) and cross-sectional dimensions (width × thickness) of each strip. They calculate cross-sectional area A in m^2. Step 2 (12 min): Learners	Before distributing equipment, say verbatim: 'This experiment is about NUMBERS, not just what you see. Your eyes told you both strips look similar — today your data will tell you the truth.' Demonstrate how to use the vernier calliper or mm ruler for cross-section in 90 seconds. Circulate every 4 minutes. At each group, ask one targeted question from this rotation: (Group round 1) 'Show me how you calculated your cross-sectional area — what units did you use?' (Group round 2) 'What is stress? Tell me in words — not just the formula.' Apply 10-second wait time. (Group round 3)	Data-table-to-graph translation (Representational Competence Building): Learners move from raw measurements → calculated quantities → graphical representation. Each translation step requires them to connect physical meaning to mathematical symbols. The side-by-side plotting of Strip A and Strip B on the same axes makes material comparison visual and precise — directly mirroring how Kenyan civil engineers compare material data sheets before selecting bridge cables.	Observable indicators: (1) Stress values are in Pascals (N/m^2) — teacher checks units on at least 3 groups' tables. (2) Strain is a dimensionless ratio — teacher asks 'what are the units of strain?' to at least 4 individual learners during circulation; correct answer = no units / pure number. (3) The stress-strain graph has labelled axes with units, a title, and two distinguishable curves. (4) Groups can point to where on their graph permanent deformation began.

Source: CK-12 🔍 Search ARES for similar readings	hang masses in 20 g increments and record the new length after each addition. They calculate extension ΔL, strain ($\Delta L/L_0$), force F = mg, and stress (F/A) in Pascals for each mass increment. They record in the printed table. Step 3 (5 min): Groups continue adding masses until one strip shows permanent deformation or snaps (with safety precautions in place). They mark on their table the row where permanent deformation began. Step 4 (3 min): Groups use their data to plot stress (y-axis) vs. strain (x-axis) on graph paper provided, drawing best-fit curves for both strips on the same axes using different colours.	'Your strain value has no units — why? What does that tell you about what strain measures?' Apply 12-second wait time. (Group round 4, near the end) 'Your two curves are separating on the graph — what does that separation mean physically?' Apply 15-second wait time. If a strip snaps safely: freeze the class for 30 seconds and say: 'Note exactly what happened — at what stress level did fracture occur? Add a mark labelled FRACTURE POINT to your graph right now.'		
Explain Phase 📹 **VIDEO:** Complex numbers with the same modulus (absolute value) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Sine Graph and Cosine Graph: Changing Amplitude (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Whole-class debrief and concept consolidation. Groups post their stress-strain graphs on the classroom wall. Learners do a 3-minute 'Gallery Walk' — each learner places a sticker dot on any graph that shows a clearly visible elastic limit point and writes one observation on a sticky note. Teacher then leads a structured whole-class discussion using a pre-drawn model stress-strain graph on the board (or projected). Learners label key features on their own graphs: (1) Linear (Hookean) region, (2) Elastic limit, (3) Yield point, (4) Ultimate Tensile Strength (UTS), (5) Fracture/Breaking point. Teacher introduces Young's Modulus as the gradient of the linear region: E = stress/strain = (F/A)/($\Delta L/L_0$). Each group calculates E for both Strip A and Strip B from the linear portion of their graph. Groups compare their E-values and discuss: 'Which rope has the higher Young's	During Gallery Walk, circulate silently and note which groups have the clearest elastic limit visible. After Gallery Walk, say: 'I noticed Group 3 and Group 7 had very clear elastic limit points. Group 3 — come tell the class how you identified it on your graph. You have 30 seconds.' Apply 12-second wait time before prompting. After group explains, say to class: 'Do you agree? Thumbs up/sideways/down.' Address any thumbs-sideways or down. When introducing E: write on the board: '$E = \sigma/\varepsilon$' then say verbatim: 'Young's Modulus is a material's resistance to being stretched per unit of strain. A HIGHER E means the material is STIFFER — it needs more stress to produce the same strain. Now — which rope had the higher E in your experiment? And what does that mean for the farmer in Kericho?' Cold-call 3 individual learners (rows 3, 6, 9) for this final	Claim-Evidence-Reasoning (CER) Framework: Learners are required to state a CLAIM ('Strip A / Strip B is safer for the bridge'), cite EVIDENCE (their calculated E-values and fracture stress), and provide REASONING ('because a higher Young's Modulus means greater stiffness and the fracture stress of Strip A is above the stress caused by the 80 kg load'). This mirrors the sensemaking practice of scientists and engineers who use material property data to make safety decisions — exactly what Kenyan structural engineers do when certifying bridges.	Observable indicators: (1) Each group's calculated E-value is in Pascals (Pa or MPa) — teacher checks at least 5 groups. (2) When cold-called, learners can state the CER triad (claim, evidence, reasoning) without prompting. (3) Learners correctly identify that a LOWER E in Strip B means it deforms more easily under the same stress — linking back to why the old Kericho rope failed. (4) Exit check: teacher asks the whole class to hold up their graph — every graph must have all 5 features labelled.

	Modulus? What does a higher E mean — stiffer or more flexible? Which rope would you use for the Kericho bridge?' The class constructs a consensus answer on the board linking E-values back to the bridge phenomenon.	question with 15-second wait time each. Write their answers verbatim on the board. Conclude with: 'Notice — we are no longer guessing. We have a NUMBER — the Young's Modulus — that tells us exactly which material is safer before a single farmer steps onto that bridge.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Complex numbers with the same modulus (absolute value) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Sine Graph and Cosine Graph: Changing Amplitude (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board (DQB) that has been running across all 6 lessons. Each learner receives two sticky notes of different colours: ORANGE = 'A question today's lesson answered' and GREEN = 'A new question today's lesson raised'. Learners write individually for 3 minutes, then place their notes in the correct DQB columns. The class reads the new green questions aloud. Learners vote (show of hands) on the top 2 unanswered questions that should be addressed in Lesson 6 (the final evidence-gathering lesson). Teacher records the top 2 questions prominently on the DQB. Learners also revisit their Lesson 5 prediction sticky notes and physically move them to a column labelled 'PREDICTION CHECKED — Was I right?' — writing a brief verdict ('My prediction was correct/incorrect because the data showed…').	Hand out sticky notes and say: 'You have three minutes — no talking. ORANGE: write one question that today's experiment answered for you, and write the answer in one sentence. GREEN: write one new question that today's data raised — something we still cannot fully explain.' Set 3-minute timer. After posting, read green questions aloud one by one. After each, ask: 'Who else had a similar question?' (show of hands). Tally. After all are read: 'The top two green questions will shape what we investigate in Lesson 6 — our final evidence session. Those two questions are your homework to think about tonight.' Point to the two winning questions and say: 'Notice — a month ago we were asking why the rope snapped. Now we are asking questions that even engineering students at University of Nairobi ask. That is the power of evidence.' Explicitly connect to driving question: 'We can now partially answer: we CAN predict which rope snaps — using Young's Modulus and the fracture stress. Write that partial answer in your science journals.'	Driving Question Board iterative revision (Metacognitive Tracking): The DQB is a living document that externalises the class's collective knowledge growth. Moving prediction notes to 'PREDICTION CHECKED' makes the learning arc visible and emotionally satisfying — learners literally see their thinking change. The green/orange colour coding separates 'knowledge consolidated' from 'productive uncertainty', which mirrors scientific practice where answering one question always generates new ones.	Observable indicators: (1) Every learner places at least one note in each colour column — teacher does visual count. (2) Orange notes contain specific vocabulary from today's lesson (stress, strain, Young's Modulus, fracture stress) — teacher reads 6 random notes. (3) Green questions are scientifically productive (e.g., 'Does temperature affect Young's Modulus of sisal?' or 'What is the E-value of steel cables used on real Kenyan bridges?') rather than off-topic. (4) All learners have updated their 'PREDICTION CHECKED' column with a written verdict.
Model Building Phase 📹 **VIDEO:** Complex	Learners open their science journals to their cumulative model of the Kericho bridge rope (started in Lesson 1 and revised in each subsequent lesson). Today's	Say verbatim: 'Every lesson we have added a new layer to our model of what happened on that Kericho bridge. Today you are adding the most powerful layer yet	Cumulative Model Revision (Progressive Modelling): Each lesson's model revision is explicitly layered on the previous one, making learning progression	Observable indicators: (1) Both ropes are shown with numerically different E-values — even if estimates, the directionality must be correct (new rope E > old rope

numbers with the same modulus (absolute value) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Sine Graph and Cosine Graph: Changing Amplitude (PLIX) Source: CK-12 🔍 Search ARES for similar readings	revision task: Add a QUANTITATIVE LAYER to the model. Learners draw or annotate two rope cross-sections side by side — one for the new rope (high E, high fracture stress) and one for the old rope (low E, low fracture stress). They label: the stress (σ) acting on the rope when the 80 kg tea-leaf sack is placed on the bridge (calculated as F/A, where F = 80 × 10 = 800 N and A is estimated from their experiment), the approximate Young's Modulus of each rope, and an arrow or annotation showing at which point each rope crosses its elastic limit and fracture point. Learners annotate: 'The old rope snapped because its fracture stress was LOWER THAN the stress applied by the 80 kg load. The new rope survived because its fracture stress was HIGHER.' Learners share their revised model with their partner and give one piece of feedback using the sentence stem: 'Your model could be stronger if you also showed…'	— actual numbers. Open your journals. You have 7 minutes.' Circulate and look for: (1) both ropes shown with different E-values labelled; (2) the 80 kg load stress annotated (accept estimates); (3) elastic limit and fracture point marked on both ropes. After 7 minutes, cold-call 2 learners (1 high-confidence, 1 still-developing — identified from earlier formative checks) to project or hold up their journal models and explain one annotation. Ask the class after each: 'What is one thing this model shows clearly? What is one thing that could be made more precise?' Apply 10-second wait time. Close with: 'In Lesson 6, we will look at how Kenyan engineers use material data sheets — with E-values and UTS values — to certify that real structures are safe. Your model today is the scientific foundation for that engineering decision.'	visible. Adding quantitative annotations (actual stress values, E-values) transforms the model from a qualitative narrative into a proto-engineering diagram. This mirrors how structural engineers at the Kenya National Highways Authority (KeNHA) annotate material property data onto structural drawings before approving bridge construction.	E). (2) The 80 kg load is converted to a stress value in Pascals and labelled on the model — teacher checks at least 6 journals. (3) The fracture point is shown only on the old rope diagram, not the new rope. (4) Peer feedback uses the sentence stem and contains at least one specific scientific suggestion.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:

1. VOCABULARY GAP: Were learners consistently distinguishing between stress (force per unit area) and strain (fractional extension)? If learners conflated the two, plan a 5-minute warm-up card-sort activity for Lesson 6 where pairs match definitions, formulas, units, and physical meaning to each term.

2. GRAPH QUALITY: How many groups produced a stress-strain graph with a clearly identifiable elastic limit? If fewer than 60% of groups achieved this, consider providing a partially pre-drawn graph template in Lesson 6 so that graphing mechanics do not consume time needed for interpretation.

3. E-VALUE ACCURACY: Were groups' calculated Young's Modulus values in a reasonable range for rubber (approximately 0.01–0.1 GPa)? Large outliers suggest errors in cross-sectional area measurement (the most common source of error). If so, build a 3-minute 'unit check' routine into future experiment phases.

4. DQB QUALITY: Were learners' green 'new question' sticky notes scientifically productive and connected to the driving question? If many notes were off-topic or trivial, introduce a sentence stem for green notes in Lesson 6: 'I still wonder whether [variable] affects [material property] because today's data showed...'

5. EQUITY CHECK: Which learners have not yet been cold-called across Lessons 1–5? Use your class register to ensure every learner is called at least once in Lesson 6. Pay particular attention to female learners in mixed-gender groups, who may be deferred to male peers during experimental work — assign rotating roles (measurer, recorder, calculator, reporter) explicitly.

6. PHENOMENON CONNECTION: Could every learner, by the end of the lesson, state in one sentence why the old Kericho rope snapped using the word 'fracture

stress' or 'Young's Modulus'? If not, the opening of Lesson 6 should include a 3-minute whole-class oral revision of this sentence before new content is introduced.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners loaded two rubber strips (representing new and old sisal rope) with increasing masses and measured their extension at each step, recording force, cross-sectional area, original length, and extension in a structured data table. They observed that Strip B (fatigued/old rope) reached permanent deformation and fracture at a lower load than Strip A, and that the two strips produced visibly different stress-strain curves when plotted on the same axes.
What did I learn?	Stress ($\sigma = F/A$, measured in Pascals) describes how much force acts per unit area inside a material. Strain ($\varepsilon = \Delta L/L_0$, dimensionless) describes how much a material deforms relative to its original length. Young's Modulus ($E = \sigma/\varepsilon$) is a material constant that quantifies stiffness — a higher E means a stiffer material that requires more stress to produce the same strain. The stress-strain graph has five key regions: linear (Hookean) region, elastic limit, yield point, Ultimate Tensile Strength (UTS), and fracture point. The old Kericho rope snapped because its fracture stress was lower than the stress applied by the 80 kg tea-leaf sack — a fact that can be determined before loading by measuring E and comparing it to the applied stress.
How does this explain the phenomenon?	The farmer's old sisal rope failed because years of use, sun exposure, and repeated loading had degraded the rope's internal fibre structure, reducing its Young's Modulus and — critically — lowering its fracture stress below the stress imposed by the 80 kg load. The new rope survived because its higher Young's Modulus and higher fracture stress meant it could bear the applied stress while remaining in the elastic (recoverable) region. By measuring stress, strain, and Young's Modulus before selecting a rope, a farmer or engineer can predict — with numbers, not guesswork — which material is safe for a given load. This is the same principle used by Kenyan civil engineers when selecting cables, tendons, and support ropes for real bridges and structures.

LESSON 6 (80 minutes): Final Explanation: Elastic Potential Energy, Elastic Limits, and Choosing the Right Material for a Safe Structure

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To consolidate all prior learning into a final scientific explanation of why the Kericho rope snapped, using elastic potential energy, Hooke's Law, and elastic limit concepts quantitatively.
Knowledge	Learners explain that elastic potential energy stored in a stretched material is given by $E_p = \frac{1}{2}kx^2$; that the elastic limit is the maximum force beyond which a material deforms permanently; and that exceeding this limit caused the older Kericho rope to fail. Learners connect force-extension graphs, spring constant values, and energy storage to safe working load decisions.
Skills	Learners calculate elastic potential energy using $E_p = \frac{1}{2}kx^2$; read and interpret force-extension graphs to identify elastic limits; apply Hooke's Law quantitatively in Kenyan engineering contexts (vehicle suspension springs, bungee cords); compare their Lesson 1 model with their final model to identify conceptual growth.
Attitudes	Learners appreciate the life-saving value of understanding material limits; demonstrate responsibility in communicating safe-load recommendations; show intellectual honesty by revising earlier predictions in light of evidence.
Key Inquiry Question	How can we use measurements of a material's spring constant and elastic limit to predict whether it will safely carry a given load — and how does this explain the failure of the older Kericho rope?
Purpose in Storyline	This is the culminating lesson of the unit. Learners return to the anchoring phenomenon — the broken Kericho bridge rope — and now possess all the conceptual tools needed to construct a complete, evidence-based explanation. They calculate the energy stored in the rope at failure, compare elastic limits, and produce a final written explanation and revised model that directly answers the driving question.
Safety Notes	No hazardous materials are used. Learners should handle spring apparatus carefully to avoid releasing stored elastic energy suddenly. Rulers and masses from earlier lessons should be returned to storage trays promptly. Remind learners not to stretch any spring beyond its marked elastic limit during any revisit activities.

B. LESSON OVERVIEW

Lesson 6 is the Final Explanation lesson for Sub-Strand 1.2. Learners open by re-examining the Kericho bridge rope phenomenon and making a refined prediction using everything they have learned over Lessons 1–5. They then engage in a structured quantitative activity calculating elastic potential energy ($E_p = \frac{1}{2}kx^2$) stored in model springs at various extensions, and identify the elastic limit on force-extension graphs from Lesson 2 data. The teacher formally introduces the elastic limit within the sensemaking discussion. Using two worked examples in Kenyan contexts — a matatu suspension spring in Nairobi traffic and a bungee cord at a Nairobi adventure park — learners apply Hooke's Law and the energy equation. In the DQB completion phase, all outstanding questions are resolved. In the final Model Building phase, learners revise their Lesson 1 initial models and place them side-by-side for a 'Model Gallery Walk', explicitly articulating the conceptual journey from naive prediction to evidence-based explanation. The lesson closes with learners writing a Final Explanation paragraph that directly answers the driving question, addressed as a formal recommendation to the Kericho farmer.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Potential energy Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Elastic Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown the original Kericho bridge rope photograph/story from Lesson 1 alongside a data card displaying: New rope spring constant k = 850 N/m; Old rope spring constant k = 210 N/m; Load applied = 80 kg (≈ 784 N); Extension of new rope = 0.92 m; Extension of old rope = 3.73 m (beyond its elastic limit of 2.1 m). Learners individually write a refined prediction in their science notebooks: 'Using what you now know, predict which rope stored more elastic potential energy at the moment of loading, and why that mattered.' They sketch a force-extension graph for each rope from memory, marking where they think the elastic limit falls on each curve. Learners share predictions with a partner using a Think-Pair-Share protocol before whole-class cold-calling.	Display the data card on the board. Say: 'We have been building our explanation for five lessons. Before I tell you anything new today, I want to see how far your thinking has come. Individually, in silence, write your best prediction right now.' Allow 4 minutes of silent individual writing. Then say: 'Turn to your partner. You have 2 minutes — share your prediction and listen to theirs. Be ready to defend your thinking.' After pair discussion, cold-call 4 learners (aim for gender balance: 2 female, 2 male) using name cards. Ask each: 'What did you predict, and what evidence from our earlier lessons supports it?' Apply 12-second WAIT TIME before accepting responses. Record key prediction phrases on the board in two columns: 'Old rope' and 'New rope'. Do not correct yet — let predictions stand for interrogation later.	Activating Prior Knowledge with Anchored Prediction — By returning to the exact phenomenon with quantitative data, learners are forced to mobilise all prior learning (spring constant, force-extension graphs, Hooke's Law, elasticity vs. plasticity) simultaneously. This creates a productive cognitive tension between their predictions and the data, priming them to notice gaps their final explanation must close.	Observable indicator: Each learner has a written prediction in their notebook with at least one piece of cited evidence from Lessons 1–5. Teacher circulates during individual writing; look specifically for learners who can correctly name the elastic limit as the threshold concept. Flag any learner who still conflates 'stretching more' with 'stronger' — these learners need targeted support in Phase 3.
Observe Phase 📹 **VIDEO:** Potential energy Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Elastic Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners work in groups of four. Each group receives: their own force-extension graph from Lesson 2 (spring data), a calculator, and a structured data table worksheet. Task 1 (10 min): Using the formula $Ep = ½kx^2$, learners calculate the elastic potential energy stored in their spring at five marked extension values from their Lesson 2 graph. They record results in a table (columns: Extension x (m) | Force F (N) | $Ep = ½kx^2$ (J)) and plot Ep against x on a new graph — observing the curve is parabolic, not linear. Task 2 (8 min): Learners are given a second data card for two model ropes (A = new Kericho rope analogue, k =	Distribute worksheets and say: 'Your job in this phase is to let the numbers tell the story. Use the formula $Ep = ½kx^2$. Start with your own spring data — you collected this evidence in Lesson 2, and now you are going to use it to understand energy storage.' Circulate actively. After 5 minutes, pause the class and ask: 'What shape is your Ep vs. x graph? Hands down — write your answer before looking at anyone else's paper.' Apply 10-second WAIT TIME, then cold-call 3 learners. Anticipated response: parabola/curve. Say: 'Why is it not a straight line? Discuss with your group for 60 seconds.' Cold-call 2	Quantitative Data Analysis with Graphical Representation — Learners generate their own Ep vs. x parabolic curve from real experimental data, making the abstract equation tangible. Comparing calculated Ep values for two ropes at the same load creates a numerical 'story' of failure: the old rope was asked to store far more energy than its structure could safely handle. This anchors the elastic limit concept in a specific, memorable calculation rather than a definition.	Observable indicator: Correct Ep calculations with units (Joules) in structured tables; parabolic curve plotted for Ep vs. x. Teacher checks at least 3 groups' Task 2 working during circulation — specifically verify that learners are computing extension via $x = F/k$ before substituting into $Ep = ½kx^2$, not using force directly. A common error is substituting F instead of x into the energy formula — address this immediately with the group before it spreads.

	850 N/m; B = old rope analogue, k = 210 N/m) and calculate Ep for each at the point where the load of 784 N is applied. They calculate the extension for each using Hooke's Law first (x = F/k), then find Ep. Task 3 (5 min): On the board, the teacher reveals the elastic limit of rope B was at x = 2.1 m, and learners calculate the maximum safe Ep rope B could store. They compare this to the Ep at the actual loading extension — observing that the applied load drove the extension far beyond the elastic limit.	groups to share. For Task 2, say: 'First use Hooke's Law to find the extension — x = F/k — then substitute into the energy formula. Show all working.' For Task 3, after revealing the elastic limit, say: 'What does it mean that rope B's actual extension of 3.73 m is nearly double its elastic limit of 2.1 m? Write one sentence in your notebook before we discuss.' WAIT 12 seconds, then cold-call 3 learners by name.		
Explain Phase 📹 **VIDEO:** Potential energy Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Elastic Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher leads a whole-class sensemaking discussion (10 min) connecting the Phase 2 calculations to the Kericho phenomenon. The teacher then formally introduces the elastic limit with a board diagram showing a force-extension graph with the linear (Hooke's Law) region, the elastic limit point, and the plastic deformation region. Learners annotate their own copy. Worked Example 1 (8 min): A matatu travelling on Thika Road has a suspension spring with k = 12,000 N/m. When four passengers (total mass 280 kg) board, calculate (a) the compression of the spring using Hooke's Law, and (b) the elastic potential energy stored in the spring. Learners attempt individually, then compare with a partner. Worked Example 2 (7 min): A bungee cord at a Nairobi adventure park on Ngong Road has k = 45 N/m and a natural length of 10 m. A person of mass 65 kg jumps. The cord begins to stretch when the person has fallen 10 m. Calculate (a) the extension when all kinetic energy has	Open the sensemaking discussion by pointing to the prediction board from Phase 1: 'Let us test your predictions with evidence.' Ask: 'Rope B stored how many joules at the moment it was loaded beyond its elastic limit? Rope A — how many joules? Which is larger? Why does that matter?' Apply 12-second WAIT TIME. Cold-call 3 learners. After responses, say: 'Here is the key idea — the elastic limit is the point of no return. Write that phrase in your notebook and draw the force-extension graph I am about to show you.' Draw the annotated graph on the board, labelling: (1) Proportional region — Hooke's Law holds; (2) Elastic limit — maximum extension before permanent deformation; (3) Plastic deformation region — material does not return to original shape; (4) Breaking point. Ask: 'Where on this graph did the old Kericho rope fail?' Cold-call 2 learners. For Worked Example 1, say: 'Attempt this alone — you have 3 minutes, show all working.' Then: 'Check with your partner. Disagree? Work it out together.' Cold-call one pair	Concept Formalisation + Worked Example Transfer — The teacher formally names and diagrams the elastic limit after learners have encountered it implicitly in data, ensuring the concept is grounded in evidence rather than delivered abstractly. The two Kenyan worked examples (matatu suspension, Nairobi bungee cord) require learners to transfer the same equations to new contexts, deepening procedural fluency while keeping the Kenyan relevance explicit. The 'bridge back' sentence forces explicit connection to the anchoring phenomenon.	Observable indicator: Learners can correctly annotate the four regions of a force-extension graph. For worked examples, correct numerical answers with full working shown: Example 1 — (a) $x = F/k = (280\times9.8)/12000 \approx 0.23$ m; (b) $Ep = ½ \times 12000 \times 0.23^2 \approx 317$ J. Example 2 — $Ep = mgh = 65\times9.8\times10 = 6370$ J; $x = \sqrt{(2Ep/k)} = \sqrt{(2\times6370/45)} \approx 16.8$ m; total length = 10 + 16.8 = 26.8 m. Teacher uses mini-whiteboards or slates for learners to hold up numerical answers — quickly scan for correct values before moving on.

	converted to elastic potential energy (assume the person has velocity ≈ 0 at maximum stretch; use $E_p = mgh$ to find energy, then solve for x), and (b) the total length of the cord at maximum stretch. Learners work in pairs. Final connect-back (5 min): Learners write a 'bridge back' sentence — 'This is similar to/different from the Kericho rope because…'	to present on the board. For Worked Example 2, say: 'This one needs two steps — energy first, then extension. Work with your partner. I will cold-call someone from every group.' Circulate and ask probing questions: 'What energy equation connects height to elastic potential energy?' 'What assumption are we making about speed at maximum stretch?' After the 'bridge back' sentences, cold-call 4 learners and record their sentences on the board under 'Connecting Back to Kericho.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Potential energy Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Elastic Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class returns to the Driving Question Board that has been built across Lessons 1–5. Each group is assigned 2–3 sticky note questions from the DQB that fall within their expertise from the unit. Groups have 6 minutes to write a concise answer (2–3 sentences, using scientific vocabulary) on a new sticky note of a different colour and attach it beside the original question. Questions not yet answered are discussed whole-class. The teacher facilitates a 'DQB Harvest' — reading each answered question and its answer aloud, asking the class to give a thumbs-up (agree), thumbs-sideways (partially agree), or thumbs-down (disagree) for each response. Disagreements are resolved through brief class discussion. The Driving Question itself — 'Why do some materials stretch and snap back while others permanently deform or break — and how can we use this knowledge to choose the right material for a safe structure in Kenya?' — is kept visible as the final target. Learners are told: 'Your final model and explanation	Point to the DQB and say: 'This board has held our questions since Lesson 1. Today we close it. Every question on this board should be answerable by someone in this room — because you have done the work.' Assign question clusters to groups. After 6 minutes, call attention and begin the 'DQB Harvest.' For each answered question, read it aloud and prompt: 'Thumbs signal — do you agree this answer is complete and accurate?' For any thumbs-sideways or thumbs-down, say: 'Who can improve this answer? Use evidence from our lessons.' Cold-call 2 learners per disputed answer. Apply 10-second WAIT TIME before accepting responses. After all questions are addressed, point to the driving question and say: 'This is the big one. We are not answering it yet — that is for your final model in Phase 5. But I want one volunteer to read it aloud, slowly, so we all hear what we are working toward.' Call on a learner who has been less vocal during the lesson.	Collaborative Knowledge Consolidation via DQB Harvest — The DQB has served as a public, evolving record of the class's growing understanding. The 'Harvest' ritual makes collective knowledge visible, creates accountability for accuracy, and gives every prior lesson equal status as a source of evidence. The thumbs-signal protocol is a low-stakes, high-engagement formative tool that surfaces misconceptions without singling out individuals.	Observable indicator: All questions on the DQB have an answer sticky note attached. Teacher specifically checks that answers to questions about elasticity vs. plasticity, Hooke's Law, and elastic limits use correct scientific vocabulary (e.g., 'proportional,' 'elastic limit,' 'spring constant,' 'permanent deformation'). Any answer using colloquial language only ('it just breaks') is flagged for revision during the harvest — the group must add a scientific term before the board is closed.

	in Phase 5 must answer this question completely.'			
Model Building Phase 📹 **VIDEO:** Potential energy Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Elastic Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their Lesson 1 initial model (drawn in their notebooks or on a stored card) and place it on the left side of a new page. They now draw their Final Model on the right side. The Final Model must include: (1) A labelled force-extension graph for both the new and old Kericho rope, with the elastic limit marked on the old rope's graph; (2) Annotations showing where Hooke's Law holds and where it breaks down; (3) Energy notation ($E_p = \frac{1}{2}kx^2$) showing the energy stored at the point of failure for the old rope; (4) A particle-level diagram (from Lesson 4) showing what happened to the molecular bonds in the old rope at the elastic limit; (5) A 'safe working load' arrow on the new rope's graph. After drawing (12 minutes), learners participate in a 'Model Gallery Walk' — models are displayed on desks, learners walk and place a small star sticker on one model that best explains the phenomenon, and write one observation on a sticky note for any model they visit. Finally, learners individually write their Final Explanation paragraph (8 minutes) — a formal written recommendation addressed to 'The Kericho Farmer' — structured as: (a) Observation of what happened; (b) Scientific explanation using elastic limit, $E_p = \frac{1}{2}kx^2$, and Hooke's Law; (c) A specific material recommendation with justification; (d) A measurable safe working load recommendation.	Distribute learners' Lesson 1 models and say: 'Place your first attempt on the left. Look at it honestly. In Lesson 1, what did you not yet know? Now build the model that shows everything you have learned.' Allow 12 minutes for model drawing with soft background silence. Circulate and prompt with: 'Have you shown the elastic limit on the graph?' 'Where is the energy equation?' 'What is happening to the fibres at the molecular level when the rope crosses its elastic limit?' Do not over-prompt — allow productive struggle. After the Gallery Walk, call the class back and ask: 'What was the most powerful thing you saw on someone else's model that you wish you had included?' Cold-call 3 learners. Then say: 'Now write your letter to the Kericho farmer. This farmer's safety depends on your advice. Use your science. Be specific — give numbers.' Allow 8 minutes. Collect Final Explanation paragraphs as a summative assessment artefact. Close the lesson by reading aloud the Driving Question one final time and saying: 'Raise your hand if you can answer this question better today than you could in Lesson 1.' Pause. 'That growth — that is what science education looks like.'	Model Comparison + Authentic Audience Final Explanation — Placing the Lesson 1 initial model directly beside the Final Model makes conceptual growth concrete and visible to the learner themselves, fulfilling the metacognitive goal of CBE. Addressing the Final Explanation to a real audience (the Kericho farmer) transforms the writing from a school exercise into a purposeful act of science communication, embedding the KICD value of responsibility. The Gallery Walk introduces peer assessment organically without formal rubric overhead.	Observable indicator: Final Model includes all five required components (labelled F-extension graphs for both ropes, elastic limit marked, Hooke's Law region annotated, $E_p = \frac{1}{2}kx^2$ notation with a value, particle-level diagram, safe working load arrow). Final Explanation paragraph addresses all four components (observation, scientific explanation with at least two named concepts, material recommendation, quantitative safe working load). Teacher uses Final Explanation paragraphs as the primary summative evidence for Specific Learning Outcomes (a) through (f) of the sub-strand. Learners who omit quantitative evidence in their paragraph are given written feedback: 'Your explanation is strong — add one calculation using $E_p = \frac{1}{2}kx^2$ or $F = kx$ to make it complete.'

D. TEACHER REFLECTION
After the lesson, reflect on the following questions and record brief notes in your professional journal: 1. CONCEPTUAL GROWTH: Compare 3–4 learners' Lesson 1 models with their Final Models. What conceptual shift is most visible? Which concept (elastic limit, $Ep = \frac{1}{2}kx^2$, particle-level explanation) proved most difficult for learners to integrate into their final model — and what would you do differently in an earlier lesson to scaffold it better? 2. QUANTITATIVE FLUENCY: During the Phase 2 Ep calculations, how many groups correctly computed $x = F/k$ before substituting into $Ep = \frac{1}{2}kx^2$ without prompting? If fewer than 50% did so independently, consider adding a 'formula unpacking' card to Lesson 3 in future iterations that explicitly distinguishes the two-step procedure. 3. DQB QUALITY: Were there any questions on the DQB that could not be answered by the class? These represent genuine gaps in the unit's learning sequence. Note them by question type (conceptual, procedural, application-level) — they may indicate a lesson is missing or underdeveloped. 4. KENYAN CONTEXT ENGAGEMENT: Did learners respond more actively to the matatu suspension spring example or the Nairobi bungee cord example? Learner engagement with contextual examples is a signal of cultural relevance — use the more engaging context as the primary example in future iterations and consider inviting a local civil engineer or Kenyan Athletics Kenya equipment manager as a guest speaker for Lesson 5 or 6. 5. GENDER EQUITY CHECK: Review your cold-call tally for this lesson. Did female and male learners receive equal opportunities to respond? If not, implement a structured name-card rotation system from the start of the unit in future cohorts to ensure equitable participation across all six lessons. 6. FINAL EXPLANATION QUALITY: Scan the collected Final Explanation paragraphs before the next class. Tally: How many included all four required components? How many cited at least one calculated value? How many addressed the farmer directly (audience awareness)? Use this data to inform your feedback strategy and to identify learners who may need additional support in the next sub-strand.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed quantitative data showing that the old Kericho rope ($k = 210$ N/m) was extended to 3.73 m under an 80 kg load — nearly double its elastic limit of 2.1 m — while the new rope ($k = 850$ N/m) extended to only 0.92 m, well within its elastic limit. Learners observed parabolic Ep vs. extension graphs from their own spring data, and compared energy stored in two model ropes at the same applied force.
What did I learn?	Learners learned that elastic potential energy stored in a stretched material is calculated using $Ep = \frac{1}{2}kx^2$; that the elastic limit is the maximum extension beyond which a material permanently deforms and cannot return to its original shape; that a higher spring constant means less extension under the same force and less energy stored; that the older Kericho rope failed because the applied load drove its extension far beyond its elastic limit, storing more energy than its molecular bonds could sustain; and that Hooke's Law and $Ep = \frac{1}{2}kx^2$ can be applied quantitatively to real Kenyan engineering contexts such as matatu suspension springs and bungee cords to determine safe working loads.
How does this explain the phenomenon?	Learners explained — using force-extension graphs, $Ep = \frac{1}{2}kx^2$ calculations, and particle-level diagrams — why the older Kericho rope snapped while the newer rope survived the same load: the older rope had a much lower spring constant due to degraded fibres, causing it to extend far beyond its elastic limit, permanently disrupting molecular bonds and ultimately causing catastrophic failure. Learners applied this explanation to recommend a safe working load for the bridge and justify the selection of a higher-k

	replacement rope, directly answering the unit's driving question.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.